

Blockchain and Earnings Management: Evidence from the Supply Chain

Donald Autore
Florida State University

Huimin (Amy) Chen¹
University of Massachusetts Lowell

Nicholas Clarke
University of Alabama in Huntsville

Jingrong Lin
University of Massachusetts Lowell

Abstract

We investigate whether corporate adoption of blockchain technology is associated with a change in firms' financial reporting behavior. On one hand, the features of blockchain technology (immutability, decentralized consensus, and real-time data sharing) can enhance data integrity, suggesting corporate blockchain adoptions may reduce earnings management. However, despite fast growth in blockchain adoptions, it remains unclear whether improved financial reporting quality or reduced accounting manipulation is a motivating factor in firms' blockchain adoption as firms vary in how they implement this new technology. On the other hand, the hype and/or increased expectations associated with blockchain adoptions, as well as the market's misperception that blockchain adoption could increase data integrity, may incentivize and provide opportunity for firms to upwardly manage earnings. We conduct our tests in the setting of the supply chain, as prior work establishes that shocks to one firm can impact linked firms through customer-supplier relationships. Our empirical evidence supports the latter prediction, as we find robust evidence that supplier firms' earnings management increases after their customers adopt blockchain. This result holds with numerous robustness tests. We provide direct evidence consistent with hype / increased expectations and reduced monitoring of supplier firms. Our findings suggest unintended consequences of blockchain adoption on financial reporting.

Keywords: Blockchain; Earnings Management; Supply Chain; Market Expectation, Monitoring

¹ Corresponding author. We thank Jason Xiao (joint editor), Wenxuan Hou (joint editor), the associate editor, two anonymous reviewers, Brad Badertscher, Heeick Choi, Lin William Cong, Bill Francis, Huiqi Gan, George Joseph, David Godsell, William Johnson, Bin Ke, Jeff McMullin, Eric Roux (Boston Blockchain Association), Thomas Shohfi, Li Sun, Qiang Wu, Yi Yang, Chi Zhang, and participants at the 2022 American Accounting Association (AAA) Annual Meeting, 2021 AAA Financial Accounting and Reporting Section (FARS) Virtual Midyear Meeting, 2021 University of Massachusetts (UMASS) Lowell Research Roundtable, and 2021 UMASS Lowell Faculty Lightning Talk for helpful suggestions. All errors are our own.

1. INTRODUCTION

“One theoretical application of blockchain is to financial reporting and this is exactly the point in time to discuss the advantages and disadvantages [...] Creating a private, real-time ledger would also, theoretically, remove the risk of earnings management from financial reporting, since all transactions are validated and recorded as they happen.”

- Campbell Harvey²

A blockchain is a shared digital ledger that tracks the transfer of asset ownership and stores sequential transactional data. Data on a blockchain are encrypted, signed, timestamped, and stored in chronological blocks that are shared among users and cannot be altered (Nakamoto 2008). The technical features of blockchains—immutability, decentralized consensus, and near-real time data sharing—are advantageous in the context of corporate financial reporting where reliability, verifiability, and transparency are of paramount importance. Using blockchains for accounting could improve transparency and reduce corporate executives’ opportunity to manipulate reported earnings (e.g., Yermack, 2017).

A growing number of firms are implementing blockchain technology into their businesses. In a survey of 1,386 senior corporate executives, 40% of respondents said they will invest \$5 million or more in blockchain technology in the coming year (Deloitte 2019). Anecdotal accounts suggest a variety of motivations, such as smart contracts or tracking and tracing assets through supply chains.³ As blockchains become more widely used to record corporate transactions, executives will have less ability to alter or manipulate the transactional data that forms the basis of their firms’ financial statements. Yermack (2017) argues that real-time accounting on the

² Quoted from a report titled “Blockchain for Financial Leaders: Opportunity vs. Reality”, published in “Financial Executives International”. Please refer to: <https://www2.deloitte.com/content/dam/Deloitte/us/Documents/financial-services/us-fsi-fei-blockchain-report-future-hr.pdf>

³ For example, the Vice President of Food Safety for Walmart, Frank Yiannas, argues that blockchain implementation can improve the firm’s ability to link foodborne outbreaks to their cause, increase food safety, and improve accountability among stakeholders in the food system. For details please refer to: <https://www.computerworld.com/article/3309656/qa-walmarts-frank-yiannas-on-the-use-of-blockchain-for-food-safety.html>

blockchain would reduce opportunities for firms to engage in accounting gimmicks and value-destroying real actions to manipulate reported earnings given the blockchain's irreversible, time-stamped transactions. Indeed, a recent symposium hosted by the American Institute of Certified Public Accountants highlights that blockchain technology is poised to play a significant role in shaping financial reporting quality (AICPA 2019).

Our goal is to provide an early investigation on whether corporate adoption of blockchain technology is associated with a change in firms' financial reporting behavior. On the one hand, the economic intuition outlined above is compelling, and predicts that corporate blockchain adoptions may reduce earnings management to the extent that a substantive part of the firm's transactional data is recorded on the blockchain.

⁴ However, it is not clear that improved financial reporting quality or reduced accounting manipulation is a motivating factor in firms' blockchain adoption. Further, despite the growth in the use of blockchains to record transactional data, blockchain adoption is not "one size fits all" as the extent to which firms report their transactional data on the blockchain varies across firms and is difficult to measure. Thus, despite the economic intuition that blockchain adoptions will reduce earnings management, whether it does so at this stage of blockchain implementation is an open empirical question.⁵

On the other hand, it is plausible that blockchain adoptions may incentivize firms to upwardly manage earnings. First, the capital market's "hype" towards blockchain adoption may pressure firms to upward manage their earnings. Specifically, announcements of blockchain

⁴ We focus on accrual earnings management rather than real earnings management because the technical features of blockchains are more likely to affect information systems than real activities.

⁵ The widely held view that blockchain adoption can reduce earnings manipulation derives from the ideal scenario that blockchain technology gains comprehensive implementation among economically linked firms and across all aspects of accounting data. Empirically, it is difficult to observe the extent to which blockchain has been applied. Blockchain application will likely expand rapidly in the coming years and warrant further investigation of its impact on earnings management.

adoptions are associated with positive initial stock price reactions, which prior literature attributes to increased expectations and/or hype (Cheng et al. 2019; Autore et al. 2020; Cahill et al. 2020). Increased expectations may pressure managers to upwardly manage earnings, while market hype may incentivize opportunistic managers to manage earnings while under the spotlight. Second, the perception that blockchain adoption could increase data integrity may change the external monitoring of firms' financial reporting. A growing line of literature shows that investors and analysts have limited capacity and rationally allocate information processing effort across firms (e.g., Kempf et al. 2017; Schmidt 2018), potentially leading to opportunistic behavior. For example, firms with distracted institutional investors are associated with value-destroying acquisitions, reduced disclosures, and more accruals-based earnings management (Kempf et al. 2017; Abramova et al. 2020; Garel et al. 2021). Motivated by this idea, we conjecture that blockchain adoptions may create a perception among market participants of better data quality and thus they may reduce their efforts to monitor these firms. If so, this may open the door for an opportunistic managerial response of upward earnings management.

We investigate the impact of blockchain adoption on earnings management in a supply chain setting. Specifically, we examine the impact on supplier firms' accrual-based earnings management after their customers adopt blockchain technology. The idea that customer adoptions of blockchain can impact supplier earnings management is premised on prior work that establishes customers and their suppliers as economically linked firms in which a shock to one firm can impact a linked firm. For example, Cohen and Frazzini (2008) use customer-supplier relationships as a proxy for economically linked firms and show that news about one firm leads to a subsequent price move in the linked firm, and Menzly and Ozbas (2010) find that stocks in economically related supplier and customer industries exhibit cross-predictability of each other's returns. An

implication, in our context, is that blockchain adoption of customer firms can have an impact on supplier firms. Moreover, using a supply chain setting is advantageous for two additional reasons. First, supply chain management is a major application of blockchain technology (Dickson 2016; Brody 2017; Wang et al. 2019). In Deloitte's blockchain survey, 54% (51%) of respondents indicated that their suppliers (customers) have asked or engaged with them about their blockchain strategy (Deloitte 2018). Cong and He (2019) suggest that blockchain-backed smart contracts could prevail along the supply chain because of their "decentralized consensus" feature, which enhances information sharing and improves the efficiency of contracting and transactions among customers and suppliers.⁶ Second, studying the suppliers of, rather than the blockchain-adopting firm itself, alleviates the endogeneity concern that adopting firms self-select into blockchain for unobserved reasons that also change their reporting behavior. We believe these benefits offset the drawbacks of using a supply chain setting, particularly that suppliers react differently than adopting firms, and the diffusion of the technology throughout the supply chain takes longer than more traditional information systems.

Our empirical analysis starts with identifying firms that adopt blockchain technology. From the Factiva database, we manually collect news articles that report corporate blockchain investments during 2015–2018.⁷ We include only observations where firms are either currently using or will imminently use blockchain for a commercial purpose (i.e., integrated into the firm's business operations), as opposed to firms in a preliminary research and development stage. From this blockchain-adopting firm sample, we then identify their supplier firms during 2013 to 2020

⁶ Internet Appendix I presents a case study of a supply chain application of a blockchain. Rapid Supplier Connect, a blockchain platform that contains both healthcare organizations' information (e.g., medical supply needs) and suppliers' information (e.g., identifications and qualifications), encrypts and shares transactional data between supplier and customer firms in near-real time.

⁷ The starting point of our search for news articles is 2008, the year in which Nakamoto (2008) published the seminal article regarding blockchain technology. The first news article about blockchain adoption was published in early 2015.

using the Factset Revere database. Our key measure of earnings management is performance-matched discretionary accruals (Kothari et al. 2005).

Our analysis reveals no evidence that blockchain adoption is associated with reduced earnings management, rather we find robust evidence that supplier firms' earnings management increases after their customers adopt blockchain. This result does not support the prediction that blockchain adoptions will reduce managers' ability to manipulate reported earnings due to enhanced data integrity, though this prediction came with the caveat that the extent to which firms report their transactional data on the blockchain is difficult to measure. This result, however, is consistent with the prediction that blockchain adoptions may incentivize firms to upwardly manage earnings due to either increased expectations/hype or reduced monitoring. On average, during the two years after its customer adopts blockchain, a supplier firm exhibits an increase in discretionary accruals corresponding to 27.9% of the standard deviation of this measure. This result holds when we use alternative measures of discretionary accruals, various combinations of fixed effects, and control for real earnings management. This result also stands up to a battery of identification and robustness tests, including tests that use propensity score matching based on both customers' and suppliers' characteristics and entropy balancing, tests that bootstrap the sample distribution, tests with alternative samples, and tests controlling for supply chain relationship materiality.

In additional tests, we provide evidence that customers' blockchain adoption is associated with a spillover of market expectations and/or hype to their suppliers. First, suppliers exhibit a significantly positive stock price reaction at the customer's blockchain adoption. This evidence supplements the findings of prior studies that report favorable stock price reactions for blockchain adopters. Second, analysts increase supplier firms' earnings estimates and make more favorable recommendations after these firms' customers adopt blockchain. These results suggest investors

and analysts upwardly revise their expectations of supplier firms, or potentially that blockchain hype impacts both investors and analysts of supplier firms.

Further, we provide evidence suggesting that market participants may reduce their effort in analyzing supplier firms after blockchain adoptions. We proxy for investor and analyst effort using (i) the number of visits to the EDGAR database (Drake et al. 2015) and (ii) news searches on Bloomberg terminals (Ben-Rephael et al. 2017). We find that supplier firms are associated with a significant decline in EDGAR visits and Bloomberg searches on their earnings announcement dates during the two years after the blockchain adoption compared to the two years before. We find that a decrease in monitoring effort, as proxied by these measures, is related to increased accrual earnings management.⁸

Finally, we test and rule out alternative explanations. We find no evidence that an increase in discretionary accruals reflects operational efficiency gains from blockchain adoption. We further find no evidence that an increase in discretionary accruals reflects more discretion in assessing cryptocurrencies.

This study contributes to burgeoning literature on the micro-economic implications of blockchain adoption. It is the first empirical study, to the best of our knowledge, to examine the effect of blockchain adoption on corporate financial reporting behavior. Prior empirical studies have largely focused on market reactions to firms' blockchain adoption announcements (Cheng et al. 2019; Autore et al. 2020; Cahill et al. 2020), and theoretical studies provide conceptual ideas about how blockchain adoptions may change corporate disclosures (Yermack 2017; Cong and He

⁸ A caveat in drawing conclusions from this association is that market participants may use other sources to analyze firms, such as XBRL and social media, and further one could argue that reducing monitoring effort due to blockchain adoption is bad governance. We cautiously conclude, based on our evidence, that supplier firms of blockchain adopters experience an increase in expectations / hype and a decrease in monitoring. These effects arguably provide an incentive and opportunity for upward earnings management, consistent with our main findings.

2019; Schmitz and Leoni 2019). Our study is timely as the Securities and Exchange Commission (SEC) is currently developing regulations on and inquiries into blockchain/distributed ledger applications through its Strategic Hub for Innovation and Financial Technology.⁹ Our results suggest that blockchain adoption may increase market expectations / hype and may also create the perception of high-quality information thereby resulting in less market discipline. Such effects can provide both the incentive and opportunity for firms to opportunistically manage earnings.

More broadly, our study is part of a large stream of literature that examines the effect of information technology (IT) on accounting information quality. IT applications such as enterprise systems, especially enterprise resource planning, are known to improve a firm's internal information system thereby improving managerial decisions and earnings forecasts (Poston and Grabski 2001; Hendricks et al. 2007; Dorantes et al. 2013). In addition, modern IT tools such as big data analytics, XBRL-based language, machine learning, and cloud computing, enhance monitoring by auditors (Zhang et al. 2015; Eulerich and Kalinichenko 2018; Ashraf et al. 2020; Bertomeu et al. 2021), which can improve reporting quality. This literature largely focuses on the implementation of IT in accounting or auditing processes to enhance financial information quality, whereas our study examines the effect of an emerging technology, blockchain, on data input and hence financial information quality. Our study indicates that a corporate blockchain does not necessarily directly apply to the accounting information system but can improve earnings quality by improving the quality of the data input of the information system, or conversely can worsen earnings quality due to managerial incentives that can counter the effect of data governance brought by modern technology. A policy implication of our study is that regulators should be aware of the interaction of managerial incentives and technological applications to enhance financial

⁹ More information is available at the SEC website: <https://www.sec.gov/finhub>

information quality. Our results also shed similar lights on the audit practice when blockchain is adopted to enhance audit quality (e.g., Cao et al. 2023).

The remainder of this paper is organized as follows. Section 2 describes the data, research design and summary statistics. Sections 3 and 4 present our main empirical tests of the association between blockchain adoption and earnings management. Section 5 rules out two alternative explanations, and Section 6 concludes.

2. EMPIRICAL PREDICTIONS

2.1 Reduced Earnings Management

Our first prediction is that blockchain adoption is associated with reduced earnings management across suppliers of adopting firms. This prediction is based on a blockchain's key technical features of immutability, decentralized consensus, and near-real time data sharing. Specifically, a blockchain records data in a sequential archive, and these transactional data are encrypted into a hexadecimal code of fixed length, called a hash code, which cannot be inverted. When new entries are recorded, the raw data are mixed with the header of the hash code of the previous entry and a timestamp to generate a new hash code for this entry, *ad infinitum* (Nakamoto 2008). This encryption strategy effectively creates a chain of data that is inherently connected (Haber and Stornetta 1990). Depending on the type of blockchain and the data-sharing protocols established, blockchain data can be shared among users in a private or semi-public form (Cong and He 2019). As users announce, verify, and share transactional data, the blockchain becomes the “distributed ledger” that maintains decentralized consensus among users.

Given these features, firms whose transactional data are recorded on a blockchain may have better corporate transparency and therefore less opportunity for managers to manipulate reported earnings (e.g., Yermack, 2017). At the extreme, with real-time financial reporting on the

blockchain, there would theoretically be no opportunity for managers to manipulate accounting figures.

However, at present, real-time financial reporting is not a reality, and despite the fast growth in firms' use of blockchains, it is hard to precisely measure the extent to which supplier firms' transactional data is recorded. Moreover, the underlying motivations for blockchain adoption appear to often be unrelated to financial reporting quality, and thus we cannot assume firms are actively implementing blockchains to reduce accounting manipulation. Overall, while this prediction of reduced earnings management has been theorized and widely discussed, the impact on earnings management may be driven by other factors.

2.2 Increased Earnings Management

Our second, and opposing prediction, is that blockchain adoption is associated with increased earnings management across suppliers of adopting firms. This prediction is based on the idea that blockchain adoptions can provide incentive for managers to upwardly manage earnings. We argue two possible ways blockchain adoptions can incentivize managerial opportunism.

2.2.1 Hype and/or Increased Expectations

The literature argues that corporate blockchain adoptions can generate hype, and relatedly, increased capital market expectations. As evidence, recent papers report that corporate blockchain investments are associated with positive initial stock price reactions (Cheng et al. 2019; Autore et al. 2020; Cahill et al. 2020). When applied over the supply chain, blockchain technology can strengthen the network between suppliers and customers through improved logistics, streamlined payment processing, and more transparent data sharing. Suppliers and customers are bonded through relationship-specific investments (Klein et al. 1978; Klein and Murphy 1997) and have closely related economic transactions, all of which can transmit the effect of economic shocks

(Costello 2020). When evaluating customer-supplier pairs, investors discern and integrate information along the supply chain. For example, Pandit et al. (2011) show that investors in supplier firms integrate earnings announcements from their customer firms into their stock price assessments. Dai et al. (2022) link customer firms' earnings announcement returns to supplier firms' short selling activities.

Thus, if market participants expect a performance boost from blockchain adopters, or if there is market hype for adopters, we should reasonably observe a spillover in expectations/hype to their supplier firms. Increased performance expectation can pressure supplier firm managers to manage earnings upward, as documented in prior studies such as Barton and Simko (2002) and Kasznik and McNichols (2002). These ideas motivate the prediction of a positive relationship between customer firms' blockchain adoptions and supplier firms' earnings management. A corollary prediction is that supplier firms experience a positive stock price reaction to the adoption and increased earnings expectations.

2.2.2 Reduced Monitoring

Blockchains are portrayed extensively in news articles, industry reports, conferences, and academic research as a transformative technology that will enhance financial reporting quality (Yermack 2017; AICPA 2019; Deloitte 2019). For example, the research firm Gartner in January 2020 predicted that blockchain can increase overall data quality by about 50% by 2023 (Egham, 2020). In practice, however, it may be difficult for market participants to discern the impact of this new technology on a specific component of accounting information. If they perceive improved financial reporting quality due to blockchain adoption, they may open the door for managerial earnings manipulation.

Specifically, with limited time and capacity for information acquisition and processing (Blankespoor et al. 2020), rational investors and analysts tend to focus on stocks of interest (Peng and Xiong 2006) and allocate more effort to processing information of firms with uncertain or complex information environment (Gibbons et al. 2021). Given the economic links between suppliers and customers, market participants may reasonably expect that the supplier firm's information quality is enhanced by the customer's adoption of blockchain. Consequently, efforts to search for firm-specific and contextual information related to the supplier firm's financial information could be reduced. For example, when projecting future earnings for suppliers, market participants may reduce the effort to verify such information with major customers, since they may perceive that the blockchain adopted along the supply chain would enhance the data integrity of the supplier firm.

There is empirical evidence consistent with these ideas, specifically that less information processing efforts are associated with more value-destroying acquisitions, reduced corporate disclosures, and more accruals-based earnings management (Kempf et al. 2017; Abramova et al. 2020; Garel et al. 2021). The literature reports ex-ante and ex-post effects of market participants' monitoring efforts on managerial reporting. Managers' ex-ante perception of monitoring will be factored into their reporting decisions: those who perceived greater investor monitoring will refrain from misreporting (Samuels et al. 2021). The ex-post detection of market participants' monitoring efforts will deter earnings management (Liu 2019; Samuels et al. 2021).

Thus, blockchain adoptions can result in reduced monitoring of supplier firms leading to opportunistic managerial behavior. The resulting prediction is that there is a positive relationship between customer firms' blockchain adoptions and supplier firms' earnings management. The

predicted underlying driver of this effect is reduced information acquisition and processing effort on the part of institutional investors and analysts.

3. DATA, RESEARCH DESIGN AND SUMMARY STATISTICS

3.1 Data and Sample Selection

3.1.1 Construction of Sample: Blockchain Adopting (Customer) Firms

To construct our sample, we search the Dow Jones Factiva global news database for news articles that contain the keyword “blockchain” and were published between October 2008 (the publication month of the original whitepaper on Bitcoin) and the end of March 2018.¹⁰ We find 47,000 news articles collectively mentioning 584 NYSE- and Nasdaq-listed stocks (excluding mutual funds and exchange-traded funds). Among these 584 stocks, 207 are reported to be investing in blockchain technology and have data—available from the Center for Research in Security Prices (CRSP) and the Compustat databases—adequate for our analyses. We use the date of the first article that mentions a firm’s blockchain investment as the date of blockchain adoption. We further screen these firms based on their stage of blockchain investment. We read each news article and categorize a firm’s blockchain investment as “advanced stage” if the news announcement explicitly reports that blockchain is or will imminently be integrated into the firm’s business operations. This leads to our final sample of 45 blockchain adopters at the advanced stage.¹¹ We further categorize the sample according to the adoption purposes in section 6.2 of this paper, and show example announcements for each category in Appendix I.

¹⁰ We follow Autore et al. (2021) in constructing our main blockchain adoption sample. Please refer to that study for a detailed description of the sample construction of blockchain adopters.

¹¹ Five of these 45 advanced stage firms will imminently adopt blockchain as opposed to already adopting. They either expect to apply the technology within a few months or recently acquired a firm that uses blockchain. In an unreported robustness test, we exclude these five firms from the baseline regression, and the results are similar to those reported.

3.1.2 Sample Construction: Supplier Firms of Blockchain Adopters

We obtain supply chain relationship data from the FactSet Revere database, including supplier–customer pairs and the start and end date of the relationship. FactSet Revere collects these data from a range of primary information sources, including annual reports, 10-K filings, 10-Q and 8-K filings, investor presentations, company websites and press releases. Merging the supply chain data with Compustat and CRSP data from 2013 to 2020 generates 126,913 supplier-years.

Panel A of Table 1 reports the supplier firm sample construction process. Panel B of Table 1 reports the sample distribution of advanced-stage blockchain-adopting customer firms by adoption years and industries. Only 13% of our sample is related to the financial industry, indicating that blockchain application has developed well beyond this industry.

[Insert Table 1]

3.2 Research Design

3.2.1 Earnings Management Measures

Our primary measure of accrual earnings management is the performance-matched discretionary accruals measure developed by Kothari et al. (2005). We first estimate the following cross-sectional model from Dechow et al. (1995) within each fiscal year and industry (SIC two-digit code):

$$\frac{TA_{i,t}}{AT_{i,t-1}} = \beta_0 + \beta_1 \frac{1}{AT_{i,t-1}} + \beta_2 \frac{\Delta REV_{i,t}}{AT_{i,t-1}} + \beta_3 \frac{PPE_{i,t}}{AT_{i,t-1}} + \epsilon_{i,t}, \quad (1)$$

where i indexes firms and t indexes fiscal years.

Next, we feed the estimated coefficients from Equation (1) into Equation (2) as follows for each firm-year observation to compute the fitted value of nondiscretionary accruals (NA_t):

$$NA_{i,t} = \widehat{\beta}_0 + \widehat{\beta}_1 \frac{1}{AT_{i,t-1}} + \widehat{\beta}_2 \frac{\Delta REV_{i,t} - \Delta REC_{i,t}}{AT_{i,t-1}} + \widehat{\beta}_3 \frac{PPE_{i,t}}{AT_{i,t-1}}. \quad (2)$$

All variables in Equations (1) and (2) are defined in Appendix II. Finally, we calculate *Accruals_Kothari*, as $DA_{i,t} - DA_{j,t}$, where j denotes a matched company with the closest return on assets (ROA) in the same fiscal year and same industry (Kothari et al. 2005). *Accruals_Kothari* is signed and constructed to be positively correlated with income-increasing earnings management. For each variable name, we use the prefix “S” or “C” to indicate whether the variable is for supplier firms (S) or customer firms (C).

Four alternative measures of earnings management are used in our baseline analyses to ensure the robustness of the results: the absolute value of performance-matched discretionary accruals (*S_Abs_Accruals_Kothari*), modified Jones Model discretionary accruals (*S_Accruals_Dechow*) (Dechow et al. 1995), performance-matched modified Jones Model discretionary working capital accruals (*S_Working_Accruals_Dechow*) (Dechow et al. 1995; Kothari et al. 2005),¹² and meet-or-beat earnings estimates (*S_MBE*). *S_Abs_Accruals_Kothari* serves as a conservative measure to verify the baseline result because unsigned accruals have lower statistical power than the signed measure *S_Accruals_Kothari* (Hribar and Nichols 2007). *S_Accruals_Dechow* is used to tease out the ROA matching effect embedded in *Accruals_Kothari*. *S_Working_Accruals_Dechow* captures the accruals related to working capital, a component significantly affected by transactions along the supply chain. *S_MBE* reflects a firm’s suspected tendency to manage earnings when it just meets or beats the earnings target. All these variables are defined in Appendix II.

3.2.2 Empirical Specification

We test the effect of customers’ blockchain adoptions on suppliers’ accrual earnings management using the following ordinary least squares (OLS) model as our baseline regression:

¹² We follow Kothari et al. (2005) and match discretionary working accruals by ROA. Alternatively, we construct nonmatched discretionary working accruals and find similar results.

$$Earnings_Management_{i,t} = \beta_0 + \beta_1 C_POST_{i,t} + \beta_2 Control\ Variables_{i,t} + Fixed\ Effects + \epsilon_{i,t}, \quad (3)$$

where $Earnings_Management_{i,t}$ is one of the five measures described in Section 3.2.1. for firm i in year t . $C_POST_{i,t}$ is an indicator variable that takes a value of one if year t is after the customers' blockchain adoption, and zero otherwise. We follow prior studies on accrual earnings management (e.g. Fang et al. 2016) and control for supplier firm fundamentals, including return on assets (S_ROA), market capitalization (S_SIZE), the book-to-market ratio (S_BM), and leverage (S_LEV). We also control for research intensity (S_RD), measured as research and development expenses scaled by net sales.¹³ We expect that research and development on technology may strengthen suppliers' integration of customers' blockchain technology and enhance the governance effect of blockchain on opportunistic behavior. All continuous variables are winsorized at the 1st and 99th percentiles. We control for various combinations of fixed effects, including firm and year fixed effects, supply chain relationship and year fixed effects, and industry and year fixed effects. All variables are defined in Appendix II.

3.3 Summary Statistics

Panel A of Table 2 presents summary statistics for our measures of earnings management and control variables. There are 1,670 supplier firm-year observations with available data to perform our baseline estimation with controls. The mean and median values of $S_Accruals_Kothari$ are 0.022 and 0.007, respectively. These values are comparable to those found in prior studies (e.g. Chen and Huang 2013). The summary statistics reported for the other control variables are also similar to those of prior studies (e.g. Fang et al. 2016).

[Insert Table 2]

¹³ We replace missing values for R_D with zero, increasing the sample size by about 300 observations in the baseline regression. In untabulated tests, we find similar results when we do not replace missing values of R_D with zeroes.

4. EVIDENCE ON EARNINGS MANAGEMENT AROUND BLOCKCHAIN ADOPTIONS

4.1 Univariate Analysis

Panel B of Table 2 reports a univariate test examining supplier firms' earnings management during the periods before and after their customer firms' adoption of blockchain technology (hereafter, "pre-adoption" and "post-adoption"). For each earnings management measure, there are significant differences across the pre- versus the post-adoption periods. For example, the mean value of *S_Accruals_Kothari* among supplier firms increases from -0.017 (pre-adoption) to 0.058 (post-adoption), and the difference is positive and statistically significant at the 5% level (t-stat = 2.60). For visual interpretation, we plot in Figure 1 discretionary accruals for each event year, from $t-3$ to $t+3$, where $t-1$ represents the 365 days immediately prior to the blockchain adoption, and $t+1$ reflects the 365 days starting immediately after the adoption date. The plot does not include "year t " because " t " represents only a single day. Discretionary accruals are relatively stable during the three pre-adoption years but increase substantially in the post-adoption years, remaining elevated through year $t+3$ in relation to pre-adoption levels.

4.2 Baseline Regression

Panel A of Table 3 reports the estimates from our baseline OLS regression based on Equation (3). The dependent variable is *S_Accruals_Kothari*, and the variable of interest is *C_POST*. Seven specifications are estimated with various inclusions of fixed effects and control variables. Specification (1) includes only the *C_POST* indicator; specification (2) adds our set of control variables; and specification (3) adds additional variables that control for customer firm characteristics (i.e., *C_ROA*, *C_SIZE*, *C_BM*, *C_LEV*, *C_RD*, and *C_Accruals_Kothari*). Standard errors are clustered by firm and year. In specification (3), each of the customer firm control variables enters insignificantly. For brevity, we do not report these estimates and exclude these

variables in all later analyses. Specifications (4) through (7) include year fixed effects, year and firm fixed effects, year and customer–supplier relationship fixed effects, and year and industry fixed effects, respectively.

[Insert Table 3]

In all seven specifications, the coefficients of *C_POST* are positive and significant (at the 1% or 5% level), indicating that supplier firms *increase* earnings management after their customer firms adopt blockchains. For example, specification (7) reports that when industry and year fixed effects are included, the coefficient of *C_POST* equals 0.165 (t-stat = 3.440). This corresponds to 27.9% of the standard deviation of discretionary accruals in our sample. This magnitude is large relative to those documented in prior earnings management studies such as Fang et al. (2016). In subsequent tests, we use specification (7) that contains industry and year fixed effects to preserve degrees of freedom.

Panel B of Table 3 reports regression estimates when *Accruals_Kothari* is replaced by alternative earnings management measures. The results are consistent with those in Panel A of Table 3. Untabulated robustness tests show similar results when we use a performance-matched measure of discretionary accruals, remove the intercept term from Equation (1), remove changes in accounts receivable in Equation (2), or include additional control variables.¹⁴ These baseline

¹⁴ Additional factors that could potentially affect our results include the capital market discipline metric and customer firm characteristics. We address these concerns by including suppliers' institutional ownership, Big N auditors, and customers' firm characteristics as control variables in the baseline regression. Untabulated results show that the coefficient of *POST* is 0.100 (t-stat = 2.672) after controlling for these additional firm characteristics. That is comparable to the coefficient of 0.165 (t-stat = 3.440) in the baseline regression. Because most of these variables have insignificant coefficients, we do not include them in the baseline regression model. Moreover, we also include controls for whether the CEO is the chair of the board of directors and whether the supplier and customer use the same auditor. These two variables enter insignificantly, and their inclusion yields similar overall results. Finally, we exclude sample adoptions in 2018 to control for the hype tied to sharp rises in cryptocurrency prices.

regression estimates point to a strong effect that blockchain adoption increases supplier firms' earnings management.^{15,16}

4.3 Identification Tests

We first perform two sets of identification tests using matching methods. The first set of tests contains difference-in-difference tests using two propensity score matched samples (Shipman et al. 2017). We construct matches using only customer firm characteristics and also a combination of customer and supplier characteristics. The second set of tests includes the difference-in-difference tests using an entropy balancing sample (Hainmueller 2012; Hainmueller and Xu 2013), which enables reweighting of the control sample using the variables in the baseline regression. In all tests we find increased earnings management after customer firms' blockchain adoption. Internet Appendix II.1 and II.2 describe the detailed matching procedures for the propensity score and entropy balancing methods, respectively. The results are presented in Internet Appendix Tables 1 and 2.

Second, we conduct a placebo test in which we replace our sample customer firms who are at an advanced stage of blockchain adoption (i.e. already using the technology) with a sample of firms at a preliminary stage of blockchain adoption. These customer firms are at the research and development stage of blockchain technology or are initiating tests such as a pilot program using blockchain. Market participants would reasonably have lower expectations and less perception of improved financial reporting quality in these blockchain adoptions, as these firms have not yet

¹⁵ A related question is whether customer firms change their accrual earnings management after adopting blockchain. In untabulated tests, we find no statistically significant change in the level of accrual earnings management from pre-adoption to post-adoption. The sample is relatively small (45 firms), and we do not attempt to draw conclusions from the result.

¹⁶ We account for the possibility that the supplier adopts blockchain before the customer's blockchain adoption. In untabulated tests we control for this effect by either (i) including a binary indicator that equals one for suppliers that adopt blockchain before their customer or (ii) removing these 66 observations. We find similar results to those reported in Table 3.

implemented blockchain. We include suppliers connected to customers at the preliminary stage of adopting blockchain and re-estimate our baseline regressions. Internet Appendix II.3 describes the findings and Internet Appendix Table 3 presents the results. There is no evidence of increased (or decreased) earnings management for preliminary stage adoptions. Taken together, the results further identify the causal relationship between customers' blockchain adoptions and suppliers' earnings management.

4.4 Robustness Tests

In addition, we conduct a battery of robustness tests for the main finding. We repeat the baseline analyses (a) based on a smaller but balanced sample,¹⁷ (b) based on a sample that contains only the supplier firms with the *first* customer blockchain adoption news and after controlling for the significance of the customer, (c) controlling for the length of the supply chain relationship, (d) controlling for real earnings management, (e) addressing whether the effect we report is temporary, and (f) accounting for the possibility of downstream blockchain integration from suppliers to customers. These robustness tests support our finding that supplier firms increase earnings management after the customer firm adopts blockchain. These tests are detailed in Internet Appendix III.

5. DIRECT EVIDENCE ON PROPOSED EXPLANATIONS FOR INCREASED EARNINGS MANAGEMENT

Section 4 presents evidence that blockchain adoptions are associated with increased earnings management. This result supports the empirical prediction (outlined in Section 2) that increased earnings management may result if supplier firms are associated with hype / increase

¹⁷ In our baseline regression sample, the supplier firm may not necessarily have data points for each of the four years surrounding the event date of blockchain adoption. For the balanced sample, each supplier firm has data for each of the four years surrounding the event date.

expectations or reduced institutional monitoring around customer blockchain adoptions. Below we provide direct tests of this prediction.

5.1 Blockchain Hype and/or Increased Expectations

In our first test, we conduct an event study of suppliers' stock price reactions to the *first* news announcement on their customer's blockchain adoption. We estimate supplier firms' cumulative abnormal returns (CARs) during narrow windows ranging from 3 to 16 days centered on its *first* customer's blockchain investment announcement. Abnormal returns are estimated using either a market model or Fama-French three-factor model (Fama and French 1993), in which the market portfolio equals the CRSP equal-weighted index.¹⁸ Panel A of Table 4 reports a mean 3-day (16-day) CAR of 0.28% (0.48%), which is statistically significant at the 1% level. This evidence suggests supplier firms experience "spillover" hype or expectations when customer firms make a blockchain investment. This finding augments the recent literature that reports favorable reactions for blockchain adopting firms (Cheng et al. 2019; Autore et al. 2020; Cahill et al. 2020). These prior studies attribute the adopter's stock price reaction to hype and/or increased expectations.

[Insert Table 4]

Our second test investigates whether analyst expectations for supplier firms change after customer firms adopt blockchain. Figure 2 plots annual averages of analyst earnings per share (EPS) estimates for supplier firms,¹⁹ and shows an increase in the analyst EPS forecast from pre-

¹⁸ Abnormal returns are calculated against benchmarks estimated using a market model or a Fama-French three-factor model over the 255 trading days ending 46 days before the event date. Market portfolio return is the CRSP equal-weighted index, and firms must have at least 22 returns during the pre-event estimation window to be included in the analysis. Significance levels are evaluated based on the Patell Z statistics (Patell 1976). We exclude supplier firm-year observations where the customer blockchain adoption is not the first of such announcements for a supplier (i.e., if a supplier has multiple customers adopting blockchain). In a robustness test, we find similar results if we exclude suppliers and customers that have earnings announcements during the 16-day event window.

¹⁹ For earnings estimates, we use the estimates for annual earnings issued in the 90 days before the fiscal year-end date and use the average estimate. For recommendations, we also use those issued in the 90 days before the fiscal

to post-adoption. Figure 3 plots the analyst recommendation scores (i.e., Strong Buy = 5, Strong Sell = 1) for supplier firms and indicates that analyst stock recommendations increase in the post-adoption period.²⁰ The figures generally display higher analyst earnings estimates and better recommendations after customer adoptions as opposed to before.

Panel B of Table 4 reports the average analyst earnings forecasts and stock recommendation scores for supplier firms in the two years before and two years after customer firms adopt blockchain. Examining four measures of market expectations for supplier firms' future performance—namely, average analyst estimates (*S_AVG_EST*), analyst recommendation scores (*S_AVG_REC*), number of upward revisions (*S_AVG_NUM_UP*) and number of downward revisions (*S_AVG_NUM_DOWN*)—we conclude that market expectations significantly increase after customer firms' blockchain adoptions. Unreported tests of medians show consistent results. Additionally, untabulated OLS regressions of analyst recommendations and earnings estimates on *POST* show statistically similar results using the controls in Jegadeesh et al. (2004) and including year fixed effects. Together, Figures 2-3 and Table 4 indicate that investors and analysts are optimistic about the earnings prospects of supplier firms after customers' blockchain adoptions.

5.2 Reduced Monitoring

We examine whether the degree of monitoring by market participants around suppliers' earnings announcements declines from before to after customers' blockchain adoptions. We construct two measures that directly reflect information processing efforts. These two measures, *S_EDGAR* and *S_BLOOMBERG*, reflect market participants' information acquisition efforts by observing their visits to a firm's SEC filings in the EDGAR database (e.g. Drake et al. 2015), or

year-end date. As a robustness test, we average estimates or recommendations over the 365 days before the fiscal year-end date. The results are similar.

²⁰ We draw similar conclusions when we plot the number of upward-revised stock recommendations and the number of downward-revised stock recommendations. For brevity, we do not report these analyses.

by examining their intensity of Bloomberg news searches, around a firm's earnings announcement date (Ben-Rephael et al. 2017).²¹

To construct S_EDGAR , we first download EDGAR filings and then calculate the number of EDGAR site visits to retrieve a firm's SEC filings.²² Following Loughran and McDonald (2017), we clean the data and obtain a dataset with statistics comparable to prior studies.²³ We define S_EDGAR as the number of visits to a supplier firm's SEC filings in the EDGAR database on the firm's earnings announcement date (Drake et al. 2015). We compare $EDGAR$ for the two years before versus the two years after customers' blockchain adoptions.²⁴ We require at least one earnings announcement in the given two-year window before and after blockchain adoption.

$S_BLOOMBERG$ is a Bloomberg news search score based on the intensity of Bloomberg news searches for a given firm on its earnings announcement date (Ben-Rephael et al. 2017). To construct $S_BLOOMBERG$, we use the Bloomberg terminal to obtain the news search score, $HEAT$. Bloomberg constructs this measure by compiling hourly counts by aggregating the number of times news articles about a particular stock are read by its terminal users and the number of times users actively search for news about a specific stock. A numerical score is generated each hour by comparing the past eight-hour average count to all hourly counts over the previous month for the

²¹ We do not choose trading-related measures (e.g., trading volume) or measures of retail investors' information processing effort (i.e., Google ticker searches), as both may capture investors' interest or limited amount of effort. Our measures are more likely to capture sophisticated investors' effort of information acquisition and interpretation, which more reasonably can have a feedback effect on managers' accounting choices. Moreover, to capture the major effort of information processing, the earnings announcement dates include both conference call dates and 10-K filing dates, whichever is earlier.

²² Since EDGAR filings search data are available only through 2017, all analysis using EDGAR visits data (Table 5) has a sample period ending in 2017.

²³ EDGAR visits with missing CIK, accession number, visitor's IP address, or date are excluded. We also remove robot visits, including those labeled as web crawler visits, index page requests, and server codes greater than or equal to 300, as well as those with 50 visits from a single IP within a single day. After cleaning, the dataset has about 24 million firm-day observations, and, on average, a given firm's EDGAR filing was accessed 15 times. These statistics are comparable with prior studies. The mean (median) daily EDGAR visits for a company in our sample is 15 (6) whereas this statistic reported in Drake et al. (2015) is 20 (8) for their sample period 2008–2011.

²⁴ As a robustness test, we also compare the number of EDGAR visits on suppliers' earnings announcement dates in the one year before versus the one year after customers' blockchain adoptions. The results are similar.

same stock. A score of 0 is assigned if the eight-hour rolling average is less than 80% of the hourly counts over the previous 30 days. In a similar fashion, a score of 1, 2, 3 or 4 is assigned if the average is between 80% and 90%, 90% and 94%, 94% and 96%, or greater than 96% of the previous 30 days' hourly counts, respectively (Ben-Rephael et al. 2017). For each sample supplier firm, we examine the change in this Bloomberg news search score on the supplier firms' earnings announcement date from before to after customers' blockchain adoptions.

Panel A of Table 5 reports the results and displays a sharp decline in supplier firm EDGAR visits on earnings announcement dates²⁵ after customers' blockchain adoptions. The average number of EDGAR visits for suppliers on their earnings announcement dates declines from about 87 in the eight quarters before the customer adopts blockchain to 32 in the eight quarters after customer adoption (difference t-stat = -4.44). The second row of this Panel shows that Bloomberg searches on suppliers' earnings announcement dates also decline after customers' blockchain adoptions in the two-year windows of pre- and post-adoption comparisons.

[Insert Table 5]

To alleviate the concern that S_EDGAR and $S_BLOOMBERG$ include only visits and searches on earnings announcement dates without considering other events or shocks that may spur investor interests, we adjust our measures to include abnormal visits and abnormal searches on *all* days during the two years before and after customer firms' blockchain adoption.²⁶ Untabulated tests show consistent results with those we report. Overall, our tests of EDGAR visits

²⁵ We use quarterly earnings announcements in the tests reported. In untabulated tests, we use annual earnings announcements and have similar findings.

²⁶ Following Ryans (2017), we construct the measure of abnormal visits by scaling the average 6 days of visits by the average 8 weeks of visits ending 1 week before the 6 days of visits. For Bloomberg searches, we follow Ben-Rephael et al. (2017) and create an indicator variable that takes the value of one if the ranking of Bloomberg news searches on the earnings announcement date is 3 or 4, and zero otherwise.

and Bloomberg searches support our argument that investor monitoring on suppliers weakens after customers' blockchain adoptions.

Lastly, we rule out the possibility that the observed reduced information processing efforts towards the supplier firms is driven by such efforts shifted to customers as they are the blockchain adaptors and therefore may attract more interest among capital market participants. To do so, we conduct the same analyses for the blockchain-adopting *customer* sample. If the effort shifting explanation is true, we should observe that information acquisition efforts increase for customer firms. However, untabulated results show that the S_EDGAR declines from 105 to 31 (difference t-stat = -2.94), $S_BLOOMBERG$ declines from 2.07 to 2.81 (difference t-stat = -4.96).

Finally, we provide a channel test in which we re-estimate our baseline specification (Column 7, Panel A, Table 3) with the inclusion of a binary indicator term for reduced monitoring and an interaction term of this indicator term multiplied by the C_POST indicator. In particular, S_DES_EDGAR and $S_DES_BLOOMBERG$ equal one if a supplier firm exhibits a decline in the two-year average of S_EDGAR or $S_BLOOMBERG$, respectively, in the post-adoption period when compared to the pre-adoption period, and zero otherwise. Panel B of Table 5 reports the results. It shows that the coefficient of the interaction term $S_DES_EDGAR * C_POST$ equals 0.183 (t-stat = 1.898),²⁷ and that of $S_DES_BLOOMBERG * C_POST$ equals 0.126 (t-stat = 1.988). These results indicate that the supplier firms with decreased monitoring have more pronounced upward earnings management after customer blockchain adoptions compared to supplier firms without decreased monitoring.

²⁷ Note that the sample size is limited because the EDGAR visit data are available only through 2017, while most blockchain adoptions occurred in 2017 and 2018. We include all blockchain adoptions that have available EDGAR visit data after adoption.

6. Additional Analysis

6.1 Ruling Out Alternative Explanations

One alternative explanation for our results is that increased discretionary accruals reflects operational efficiency gains from adopting blockchain, because the technology accelerates transactions and hence turnovers of working capital, the major component of accruals.

To test this explanation, we examine whether blockchain adoption brings changes to operating efficiency that leads to increases in earnings management. Following prior studies (e.g. Barton and Simko 2002; Cohen and Zarowin 2010), we construct five measures of operation efficiency: days of the operating cycle, inventory turnover rate, accounts receivable turnover rate, asset turnover rate, and accounts payable turnover rate. We then compare the average of each variable during the two years after blockchain adoption versus the two years before adoption and create indicator variables of whether operational efficiency improves. S_CYC takes the value of one if suppliers have fewer operating cycle days after blockchain adoption, and zero otherwise. S_INV , S_REC , S_AT , and S_AP each take the value of one if suppliers have an increased inventory, receivable, asset, and payable turnover rate after blockchain adoption, respectively, and zero otherwise.

In Table 6, we report the results of estimations in which our baseline regression is augmented with these indicator variables and their interaction with C_POST . The estimates show that the coefficient of the interaction term in each specification is not statistically significant. This evidence provides no support for the alternative explanation that the increase in discretionary accruals is driven by gains in operational efficiency.

[Insert Table 6]

6.2 The Purpose of Blockchain Adoption

The reason behind the customer's adoption of blockchain could differentially affect the degree of market expectation and discipline, which in turn could affect suppliers' reporting behaviors. We categorize the blockchain adopting firms into groups by their stated purpose for adopting blockchain: a "Financial" category to provide financial services (cryptocurrency or fintech), and "Nonfinancial" category for all other blockchain adoption purposes. The "Financial" category mainly includes blockchain adoptions for cryptocurrency or fintech. The cryptocurrency category comprises firms that apply blockchain on currency exchanges, and the fintech category encompasses companies that provide supporting financial services, including financial data offering, remittance service, and future trading. By their adoption purposes, 13 firms are categorized as "Financial," if blockchain is used to provide financial services (i.e., cryptocurrency or Fintech industry), and the other firms are categorized as "Nonfinancial". Among "Nonfinancial" blockchain adoptions, there are "Advantage" and "Information" purposes. The "Advantage" category includes firms adopting blockchain for a general advantage such as commitment to online innovations or gaining competitive advantage in the blockchain industry. Firms can also adopt blockchain for other advantages including specific operational purposes, including inventory tracking (e.g. molecular tagging, crop tracking, metal refinery tracing) and supply chain management (e.g. drug transportation verification). The "Information" category contains firms that use blockchain to improve information quality, where the news articles mention data or information quality purposes related to the blockchain adoption.

We repeat the baseline analyses based on these four subsamples. Internet Appendix Table 4 presents the results of tests that follow our baseline estimations using these four subsamples. Consistent with our expectation, the results show that the coefficients of *C_POST* are significantly

positive in the *Nonfinancial* and *Advantage* groups, but not in the *Financial* and *Information* groups. The insignificant result of the *Financial* subsample mitigates the concern that the market hype of cryptocurrency can be one driver of our result. The insignificant result of *Information* subsample provides weak support for the competing argument that blockchain mitigates managerial opportunism and reduces earnings management. It appears that firms adopt blockchain broadly for other purposes and can be driven by over hyped market expectation or diminished market discipline. However, when firms adopt blockchain for information quality purposes, earnings management does not significantly increase, suggesting there is a weak mitigating effect of blockchain on managerial opportunism or that managers avoid earnings management when their firms are committed to enhancing information quality.

7. CONCLUSION

Blockchain is among the most disruptive technologies in today's world. The application of blockchain for business purposes is widely supported by corporate executives according to the 2019 Deloitte blockchain survey (Deloitte 2019). Of 1,386 survey respondents, 40% indicated that their companies are interested in spending \$5 million or more on blockchain technology. About 95% of respondents indicated using blockchain in smart contracts, and 43% (40%) of respondents expected to use blockchain technology for data validation (data access/sharing). These applications of blockchain potentially will change how data are stored and accessed in the future and could have profound benefits for the supply chain. As such, how the application of blockchain might affect financial reporting—a process that significantly relies on the verifiability and transparency of transactional data—is of great interest to practitioners and regulators.

We test the impact of a customer firm's blockchain adoption on its suppliers' earnings management activities. We find that customers' adoption of blockchain is associated with suppliers' increased earnings management. This result is consistent with two non-mutually exclusive explanations. The first is that blockchain adoptions lead to hype and/or elevated expectations among capital market participants. The second explanation is reduced monitoring of blockchain-linked firms on the part of investors and analysts due to the perception that blockchain will enhance the quality of financial data. Our findings are consistent with both explanations, as each can have a feedback effect on managers' accounting choices by providing the incentive and opportunity to manipulate earnings. We rule out that our results are driven by gains in operational efficiency (due to blockchain) that accelerate transactions and working capital turnover. Using several proxies, we find no evidence that changes in earnings management around blockchain adoptions are linked to concurrent changes in operational efficiency.

Our study has several limitations. First, we study a relatively short sample period and a small number of customer firms that are considered early adopters of blockchain (Stratopoulos et al. 2020). Second, real-time financial reporting or at least reduced accounting manipulation may not be a motivating factor in firms' blockchain adoption, thus the impact of blockchain adoption is hard to predict ex-ante. Third, firms use blockchains for a variety of reasons and therefore the extent to which firms report their transactional data on the blockchain varies, and this variation is difficult to measure. Fourth, it is difficult to measure monitoring effort. Over the sample period of our study, the number of ways market participants can monitor firms has increased (particularly with the growth in social media for reporting financial data) and therefore the extent of monitoring by market participants is difficult to measure. In this light, we cautiously conclude that our study demonstrates potential unintended consequences of blockchain adoption on financial reporting.

We hope this work provides a foundation for future studies that are less affected by these limitations as blockchain technology becomes more widely adopted.

References

- Abramova, I., J. E. Core, and A. Sutherland. 2020. Institutional Investor Attention and Firm Disclosure. *The Accounting Review* 95 (6):1-21.
- AICPA. 2019. 2019 Blockchain Symposium: Experts' Insights Indicate Growing Use Cases and Value for the Technology. Available at: <https://www.cpa.com/sites/cpa/files/2019-12/2019-blockchain-symposium-report.pdf>.
- Ashraf, M., P. N. Michas, and D. Russomanno. 2020. The Impact of Audit Committee Information Technology Expertise on the Reliability and Timeliness of Financial Reporting. *The Accounting Review* 95 (5):23-56.
- Autore, D., M. N. Clarke, and D. Jiang. 2020. Blockchain Speculation or Value Creation? Evidence from Corporate Investments. *Financial Management* Forthcoming.
- Barton, J., and P. J. Simko. 2002. The Balance Sheet as an Earnings Management Constraint. *The Accounting Review* 77 (s-1):1-27.
- Ben-Rephael, A., Z. Da, and R. D. Israelsen. 2017. It Depends on Where You Search: Institutional Investor Attention and Underreaction to News. *The Review of Financial Studies* 30 (9):3009-3047.
- Bertomeu, J., E. Cheynel, E. Floyd, and W. Pan. 2021. Using Machine Learning to Detect Misstatements. *Review of Accounting Studies* 26 (2):468-519.
- Blankespoor, E., E. deHaan, and I. Marinovic. 2020. Disclosure Processing Costs, Investors' Information Choice, and Equity Market Outcomes: A Review. *Journal of Accounting and Economics* 70 (2-3).
- Brody, P. 2017. How Blockchain Is Revolutionizing Supply Chain Management. *Ernst & Young Digitalist Magazine* Available at: [https://www.ey.com/Publication/vwLUAssets/ey-blockchain-and-the-supply-chain-three/\\$FILE/ey-blockchain-and-the-supply-chain-three.pdf](https://www.ey.com/Publication/vwLUAssets/ey-blockchain-and-the-supply-chain-three/$FILE/ey-blockchain-and-the-supply-chain-three.pdf) (last access Aug 28, 2020):1-7.
- Cahill, D., D. G. Baur, Z. Liu, and J. W. Yang. 2020. I Am a Blockchain Too: How Does the Market Respond to Companies' Interest in Blockchain? *Journal of Banking & Finance* 113:105740.
- Cao, S., L. W. Cong, and B. Yang. 2023. Financial Reporting and Blockchains: Audit Pricing, Misstatements, and Regulation. *Working paper, available at SSRN*.
- Chen, S.-S., and C.-W. Huang. 2013. The Sarbanes-Oxley Act, Earnings Management, and Post-Buyback Performance of Open-Market Repurchasing Firms. *Journal of Financial and Quantitative Analysis*:1847-1876.
- Cheng, S. F., G. De Franco, H. Jiang, and P. Lin. 2019. Riding the Blockchain Mania: Public Firms' Speculative 8-K Disclosures. *Management Science* 65 (12):5901-5913.

- Cohen, D. A., and P. Zarowin. 2010. Accrual-Based and Real Earnings Management Activities around Seasoned Equity Offerings. *Journal of Accounting and Economics* 50 (1):2-19.
- Cohen, L., and A. Frazzini. 2008. Economic Links and Predictable Returns. *The Journal of Finance* 63 (4):1977-2011.
- Cong, L. W., and Z. He. 2019. Blockchain Disruption and Smart Contracts. *The Review of Financial Studies* 32 (5):1754-1797.
- Costello, A. M. 2020. Credit Market Disruptions and Liquidity Spillover Effects in the Supply Chain. *Journal of Political Economy* 128 (9):000-000.
- Dai, R., L. Ng, and N. Zaiats. 2022. Short Seller Attention. *Journal of Corporate Finance* 72:102149.
- Dechow, P. M., R. G. Sloan, and A. P. Sweeney. 1995. Detecting Earnings Management. *Accounting review*:193-225.
- Deloitte. 2018. Deloitte's 2018 Global Blockchain Survey. <https://www2.deloitte.com/us/en/insights/topics/understanding-blockchain-potential/global-blockchain-survey.html> (last access: Mar 15, 2020).
- . 2019. Deloitte's 2019 Global Blockchain Survey. <https://www2.deloitte.com/us/en/insights/topics/understanding-blockchain-potential/global-blockchain-survey.html> (last access: Mar 15, 2020).
- Dickson, B. 2016. Blockchain Has the Potential to Revolutionize the Supply Chain. *Tech Crunch* 25.
- Dorantes, C. A., C. Li, G. F. Peters, and V. J. Richardson. 2013. The Effect of Enterprise Systems Implementation on the Firm Information Environment. *Contemporary Accounting Research* 30 (4):1427-1461.
- Drake, M. S., D. T. Roulstone, and J. R. Thornock. 2015. The Determinants and Consequences of Information Acquisition Via Edgar. *Contemporary Accounting Research* 32 (3):1128-1161.
- Eulerich, M., and A. Kalinichenko. 2018. The Current State and Future Directions of Continuous Auditing Research: An Analysis of the Existing Literature. *Journal of Information Systems* 32 (3):31-51.
- Fama, E. F., and K. R. French. 1993. Common Risk Factors in the Returns on Stocks and Bonds. *Journal of Financial Economics* 33 (1):3-56.
- Fang, V. W., A. H. Huang, and J. M. Karpoff. 2016. Short Selling and Earnings Management: A Controlled Experiment. *The Journal of Finance* 71 (3):1251-1294.
- Garel, A., J. M. Martin-Flores, A. Petit-Romec, and A. Scott. 2021. Institutional Investor Distraction and Earnings Management. *Journal of Corporate Finance* 66:101801.

- Gibbons, B., P. Iliev, and J. Kalodimos. 2021. Analyst Information Acquisition Via Edgar. *Management Science* 67 (2):769-793.
- Haber, S., and W. Stornetta. 1990. How to Time-Stamp a Digital Document. Paper read at Conference on the Theory and Application of Cryptography.
- Hainmueller, J. 2012. Entropy Balancing for Causal Effects: A Multivariate Reweighting Method to Produce Balanced Samples in Observational Studies. *Political analysis*:25-46.
- Hainmueller, J., and Y. Xu. 2013. Ebalance: A Stata Package for Entropy Balancing. *Journal of Statistical Software* 54 (7).
- Hendricks, K. B., V. R. Singhal, and J. K. Stratman. 2007. The Impact of Enterprise Systems on Corporate Performance: A Study of ERP, Scm, and Crm System Implementations. *Journal of Operations Management* 25 (1):65-82.
- Hribar, P., and C. Nichols. 2007. The Use of Unsigned Earnings Quality Measures in Tests of Earnings Management. *Journal of Accounting Research* 45 (5):1017-1053.
- Hui, K. W., S. Klasa, and P. E. Yeung. 2012. Corporate Suppliers and Customers and Accounting Conservatism. *Journal of accounting and Economics* 53 (1-2):115-135.
- Jegadeesh, N., J. Kim, S. D. Krische, and C. M. Lee. 2004. Analyzing the Analysts: When Do Recommendations Add Value? *The Journal of Finance* 59 (3):1083-1124.
- Jones, J. J. 1991. Earnings Management During Import Relief Investigations. *Journal of Accounting Research* 29 (2):193-228.
- Kasznik, R., and M. F. McNichols. 2002. Does Meeting Earnings Expectations Matter? Evidence from Analyst Forecast Revisions and Share Prices. *Journal of Accounting Research* 40 (3):727-759.
- Kempf, E., A. Manconi, and O. Spalt. 2017. Distracted Shareholders and Corporate Actions. *The Review of Financial Studies* 30 (5):1660-1695.
- Klein, B., R. G. Crawford, and A. A. Alchian. 1978. Vertical Integration, Appropriable Rents, and the Competitive Contracting Process. *The journal of Law and Economics* 21 (2):297-326.
- Klein, B., and K. M. Murphy. 1997. Vertical Integration as a Self-Enforcing Contractual Arrangement. *The American Economic Review* 87 (2):415-420.
- Kothari, S. P., A. J. Leone, and C. E. Wasley. 2005. Performance Matched Discretionary Accrual Measures. *Journal of accounting and Economics* 39 (1):163-197.
- Liu, Y. 2019. Information Acquisition Costs and Misreporting: Evidence from the Implementation of Edgar.

- Loughran, T., and B. McDonald. 2017. The Use of Edgar Filings by Investors. *Journal of Behavioral Finance* 18 (2):231-248.
- Menzly, L., and O. Ozbas. 2010. Market Segmentation and Cross-Predictability of Returns. *The Journal of Finance* 65 (4):1555-1580.
- Nakamoto, S. 2008. Bitcoin: A Peer-to-Peer Electronic Cash System.(2008).
- Pandit, S., C. E. Wasley, and T. Zach. 2011. Information Externalities Along the Supply Chain: The Economic Determinants of Suppliers' Stock Price Reaction to Their Customers' Earnings Announcements. *Contemporary Accounting Research* 28 (4):1304-1343.
- Patell, J. M. 1976. Corporate Forecasts of Earnings Per Share and Stock Price Behavior: Empirical Test. *Journal of Accounting Research* 14 (2):246-276.
- Peng, L., and W. Xiong. 2006. Investor Attention, Overconfidence and Category Learning. *Journal of Financial Economics* 80 (3):563-602.
- Poston, R., and S. Grabski. 2001. Financial Impacts of Enterprise Resource Planning Implementations. *International Journal of Accounting Information Systems* 2 (4):271-294.
- Roychowdhury, S. 2006. Earnings Management through Real Activities Manipulation. *Journal of Accounting and Economics* 42 (3):335-370.
- Ryans, J. 2017. Using the Edgar Log File Data Set. *Available at SSRN 2913612*.
- Samuels, D., D. J. Taylor, and R. E. Verrecchia. 2021. The Economics of Misreporting and the Role of Public Scrutiny. *Journal of accounting and Economics* 71 (1):101340.
- Schmidt, D. 2018. Distracted Institutional Investors. *Journal of Financial and Quantitative Analysis (forthcoming)* 54 (6):2453-2491.
- Schmitz, J., and G. Leoni. 2019. Accounting and Auditing at the Time of Blockchain Technology: A Research Agenda. *Australian Accounting Review* 29 (2):331-342.
- Shipman, J. E., Q. T. Swanquist, and R. L. Whited. 2017. Propensity Score Matching in Accounting Research. *The Accounting Review* 92 (1):213-244.
- Stratopoulos, T. C., V. X. Wang, and J. Ye. 2020. Blockchain Technology Adoption. *Available at SSRN 3188470*.
- Wang, Y., J. H. Han, and P. Beynon-Davies. 2019. Understanding Blockchain Technology for Future Supply Chains: A Systematic Literature Review and Research Agenda. *Supply Chain Management: An International Journal*.
- Yermack, D. 2017. Corporate Governance and Blockchains. *Review of Finance* 21 (1):7-31.

- Zang, A. Y. 2012. Evidence on the Trade-Off between Real Activities Manipulation and Accrual-Based Earnings Management. *The Accounting Review* 87 (2):675-703.
- Zhang, J., X. Yang, and D. Appelbaum. 2015. Toward Effective Big Data Analysis in Continuous Auditing. *Accounting Horizons* 29 (2):469-476.

Appendix I. Example Announcements of Blockchain Adoption

This appendix gives examples of how firms are categorized based on the purpose of blockchain adoption.

Purpose	Example announcements	Business Nature
Financial – Cryptocurrency	“Bitcoin futures trading will start this coming weekend, and Riot Blockchain (RIOT) scored an investment for a digital currency exchange.”	Security & commodity exchanges
Financial – Fintech industry	“...announced launch of the Cryptocurrency Data Feed. ...offers real-time cryptocurrency information and will initially include data from more than 15 cryptocurrency exchange venues globally.”	Operator of global exchanges and clearing houses & provider of data and listings services
NonFinancial – Strategic Advantage	“Unblock is tasked with integrating Blockchain technologies into cross-market system.” “The establishment of Unblock signals we are ready to invest in order to play a leading role in the blockchain industry.”	Mobile platform
NonFinancial – Operational Advantage	1) “...use blockchain technology with its molecular tagging in the authentication of physical goods”; 2) “...will start using the blockchain, a new, shared database technology that can’t easily be changed or erased. A technician taking a specimen can attach data such as collection time, a lab researcher can add reports and all will be immediately visible to anyone with access.”	1) Identification of counterfeit consumer goods; 2) Mining
NonFinancial – Operational Advantage	“Blockchain Data Integrity is now commercially available on ... platform for building industrial applications. It provides fully auditable, compliant and trustworthy data.”	Networking and telecommunications

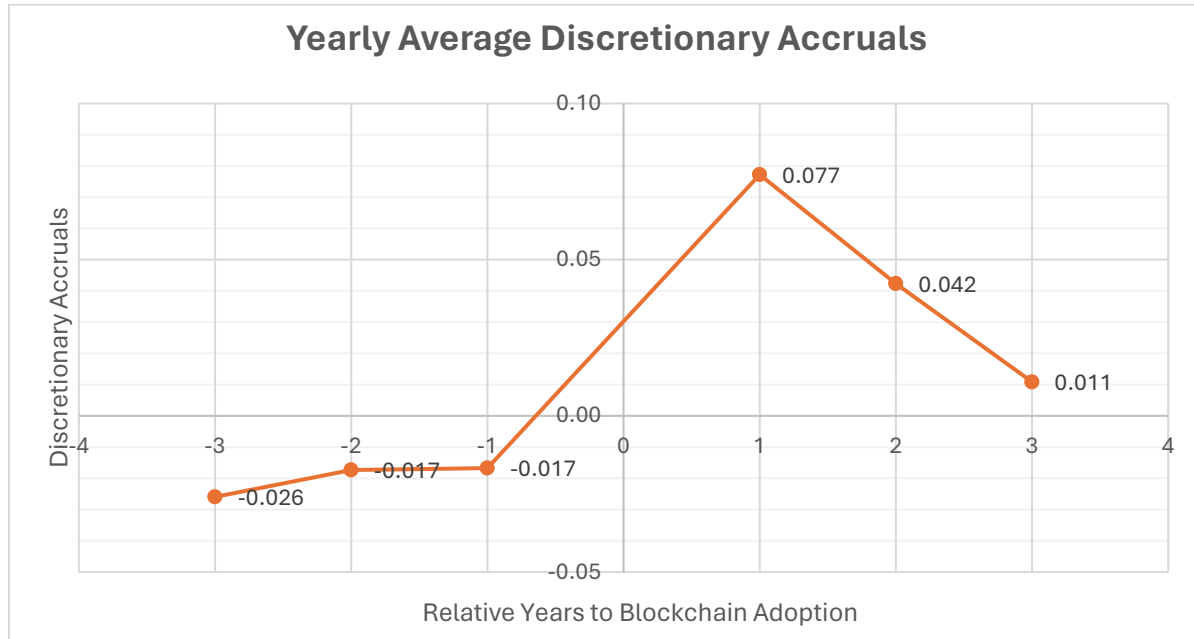
Appendix II. Variable Descriptions

Dependent Variables	
<i>S_Accruals_Kothari</i>	Kothari et al. (2005) performance-matched discretionary accruals. We follow equation (1) and (2) to construct the measure: $\frac{TA_{i,t}}{AT_{i,t-1}} = \beta_0 + \beta_1 \frac{1}{AT_{i,t-1}} + \beta_2 \frac{\Delta REV_{i,t}}{AT_{i,t-1}} + \beta_3 \frac{PPE_{i,t}}{AT_{i,t-1}} + \epsilon_{i,t} \quad (1)$ $NA_{i,t} = \widehat{\beta}_0 + \widehat{\beta}_1 \frac{1}{AT_{i,t-1}} + \widehat{\beta}_2 \frac{\Delta REV_{i,t} - \Delta REC_{i,t}}{AT_{i,t-1}} + \widehat{\beta}_3 \frac{PPE_{i,t}}{AT_{i,t-1}} \quad (2)$ Total accruals TA_t are computed as earnings before extraordinary items and discontinued operations minus operating cash flows for fiscal year t; AT_{t-1} is total assets at the end of year $t-1$; ΔREV_t is the change in sales revenue from year $t-1$ to t; and PPE_t is the gross value of property, plant, and equipment at the end of year t. We require at least 10 observations to perform each cross-sectional estimation. ΔREC_t is the change in accounts receivable from year $t-1$ to t. Firm-year discretionary accruals (DA_t) is calculated as $DA_{i,t} = (TA_{i,t} / AT_{i,t-1}) - NA_{i,t}$. The prefix “S_” indicates the variable is for suppliers.
<i>S_Abs_Accruals_Kothari</i>	Absolute value of <i>Accruals_Kothari</i> .
<i>S_Accruals_Dechow</i>	Dechow et al. (1995) modified Jones model (Jones 1991) for discretionary accruals.
<i>S_Working_Accruals_Dechow</i>	A model by Dechow et al. (1995) based on working capital.
<i>S_MBE</i>	Meet-or-beat analyst forecast estimate indicator variable, indicating whether a firm is suspected of engaging in meeting/beating earnings target behavior (e.g., (e.g. Roychowdhury 2006; Zang 2012). MBE equals one if the observation is either (a) firm-years with earnings before extraordinary items over lagged assets between 0 and 0.005 or (b) firm-years with actual earnings per share exceeding the recent analyst forecast consensus before the fiscal year-end by less than 0.01.
Variables of Interest	
<i>C_POST</i>	An indicator variable that takes a value of one if year t is after the supplier firm’s customer firm adopts blockchain technology, and zero otherwise. The prefix “C_” indicates the variable is for customers.
Control Variables	
<i>S_ROA</i>	Return on assets, estimated as earnings before income tax divided by lagged total assets (EBITDA/ AT_{t-1}).
<i>S_SIZE</i>	Firm size = natural logarithm of market capitalization.
<i>S_BM</i>	Book-to-market ratio, measured as the book value of common equity divided by the market value of equity, which

	is the closing annual price multiplied by outstanding shares $CEQ/(PRCC \cdot F \cdot CSHO)$.
<i>S_LEV</i>	Leverage, measured as long-term debt divided by total assets (DLTT/AT).
<i>S_RD</i>	Research and development intensity, measured as research and development expenses divided by sales (XRD/SALE).
Other Variables	
<i>S_AVG_EST</i>	Average analyst earnings estimate over the two-year period before or after blockchain adoption.
<i>S_AVG_REC</i>	Average analyst recommendation scores over the two-year period before or after blockchain adoption. Schemes of <i>AVG_REC</i> reverses the order of the recommendations in I/B/E/S (i.e., <i>AVG_REC</i> equals the I/B/E/S score subtracted by 6, resulted in 5–“strong buy,” 4–“buy,” 3–“hold,” 2–“sell,” 1–“strong sell”). Higher value of <i>AVG_REC</i> indicates more favorable recommendations.
<i>S_AVG_NUM_UP</i>	Average number of analyst upward revision in recommendations over the two-year period before or after block chain adoption.
<i>S_AVG_NUM_DOWN</i>	Average number of analyst downward revision in recommendations over the two-year period before or after block chain adoption.
<i>S_EDGAR</i>	The number of visits to a supplier firm’s SEC filings in the EDGAR database on the firm’s earnings announcement date (Drake et al. 2015).
<i>S_BLOOMBERG</i>	A Bloomberg news search score based on the intensity of Bloomberg news searches for a given supplier firm on its earnings announcement date (Ben-Rephael et al. 2017).
<i>S_DES_EDGAR</i>	An indicator variable that takes a value of one if a supplier has fewer EDGAR visits during the two years after the customers’ blockchain adoption than during the two years before blockchain adoption.
<i>S_DES_BLOOMBERG</i>	An indicator variable that takes a value of one if a supplier has fewer Bloomberg searches during the two years after the customers’ blockchain adoption than during the two years before blockchain adoption.
<i>S_INS_OWN</i>	Institutional investors’ ownership.
<i>S_ACCRUAL_LAG1</i>	One-year-lagged Kothari et al. (2005) performance-matched discretionary accruals.
<i>C_ROA</i>	Customer firms’ return on asset (<i>ROA</i>).
<i>C_LN(MV)</i>	Customer firms’ natural logarithm of market value (<i>MV</i>).
<i>C_LEV</i>	Customer firms’ leverage (<i>LEV</i>).
<i>C_RD</i>	Customer firms’ research and development intensity (<i>RD</i>).

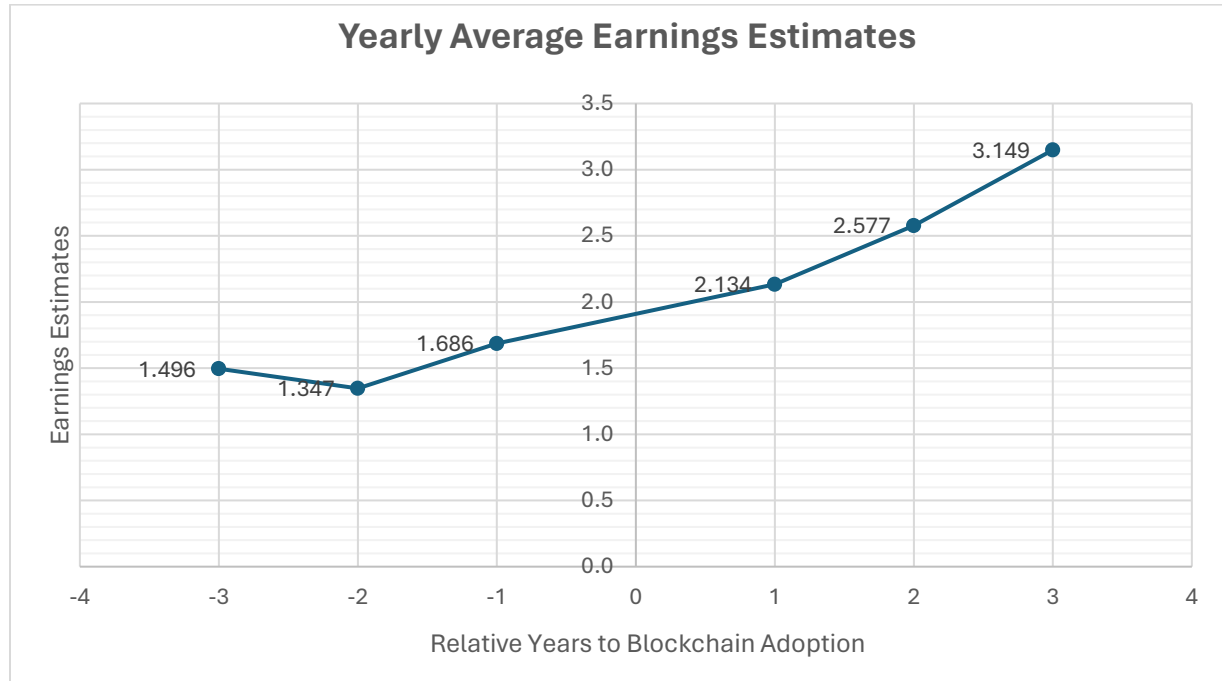
<i>C_CAPX_G</i>	Customer firms' capital market expenditure growth rate, measured as the percentage change in a firm's capital investments (CAPX) from year t to $t+1$.
<i>C_HHI</i>	Customer firms' industry Herfindahl-Hirschman Index (HHI).
<i>C_ANN_DAYS</i>	Customer firms' earnings announcement speed, measured as the number of days between the firm's fiscal year-end date and earnings announcement date.
<i>C_ACCRUALS</i>	Customer firms' Kothari et al. (2005) performance-matched discretionary accruals.
<i>S_SUP_DAY</i>	Number of days that the supplier and customer firms are in the supply chain relationship.

Figure 1: Yearly Distribution of Discretionary Accruals



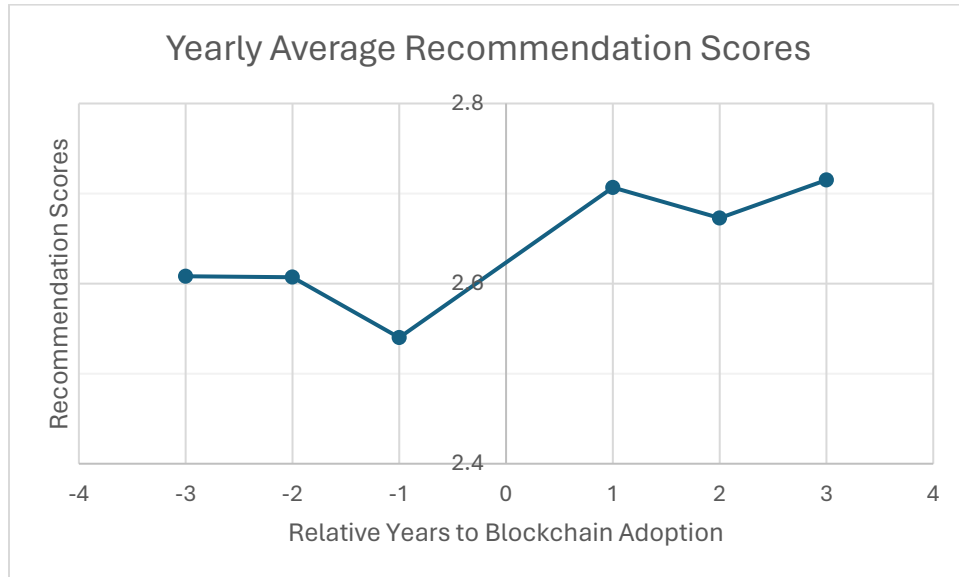
This figure plots trends in suppliers' average discretionary accruals over event years relative to customers' blockchain adoption date from Year -3 to Year +3. All variables are defined in Appendix II.

Figure 2: Yearly Distribution of Earnings Estimates



This figure plots trends in suppliers' average earnings per share (EPS) estimates over event years relative to customers' blockchain adoption date from Year -3 to Year +3. All variables are defined in Appendix II.

Figure 3: Yearly Distribution of Analyst Recommendations



This figure plots trends in suppliers' actual earnings per share over event years relative to customers' blockchain adoption date from Year -3 to Year +3. The average recommendation scores are calculated using the scheme as follows: 5—"strong buy," 4—"buy," 3—"hold," 2—"sell," 1—"strong sell".

Table 1: Sample Description

This table describes our sample. Panel A reports the sample selection process, and Panel B reports the sample distribution of blockchain-adopting customer firms.

Panel A: Sample Selection

	Supplier-years
Supplier-customer pair-years from 2013 to 2020 after merging datasets from Factset Revere, CRSP, and Compustat	126,913
Merge with blockchain customer firms at the advanced stage	20,011
Merge with Compustat data and the discretionary accruals measure	7,219
Limit the sample to two years before and after customers' blockchain adoptions	1,670

Panel B: Blockchain Sample Distribution

<u>Yearly distribution:</u>	
2015	7%
2016	16%
2017	49%
2018	29%
<u>Industry distribution:</u>	
Mining	10%
Manufacturing	29%
Transportation, Communications, Electric, Gas, and Sanitary Services	6%
Trade	10%
Finance, Insurance, and Real Estate	13%
Services	29%
Public Administration	3%

Table 2: Summary Statistics and Univariate Tests

Panel A presents summary statistics for our measures of earnings management and controls. Panel B presents univariate tests comparing multiple measures of earnings management before and after customers adopt blockchain. Detailed definitions of all variables are provided in Appendix II. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Panel A: Summary Statistics

	<i>N</i>	<i>Mean</i>	<i>Std</i>	<i>25th</i>	<i>Median</i>	<i>75th</i>
<i>S_Accruals_Kothari</i>	1,670	0.022	0.591	-0.105	0.007	0.119
<i>S_Abs_Accruals_Kothari</i>	1,670	0.312	0.652	0.045	0.113	0.272
<i>S_Accruals_Dechow</i>	1,670	0.083	0.474	-0.066	0.032	0.237
<i>S_Working_Accruals_Dechow</i>	1,670	-0.010	0.098	-0.061	-0.003	0.032
<i>S_MBE</i>	1,670	0.073	0.261	0.000	0.000	0.000
<i>S_ROA</i>	1,670	0.070	0.162	0.034	0.099	0.147
<i>S_SIZE</i>	1,670	7.642	2.481	5.921	7.660	9.267
<i>S_BM</i>	1,670	0.436	0.469	0.191	0.344	0.593
<i>S_LEV</i>	1,670	0.196	0.194	0.009	0.163	0.294
<i>S_RD</i>	1,670	0.148	0.252	0.007	0.077	0.185

Panel B: Univariate Tests

	Pre-adoption			Post-adoption				t-stat
	N	Mean	Std	N	Mean	Std	Mean Difference	
<i>S_Accruals_Kothari</i>	800	-0.017	0.454	870	0.058	0.691	0.075**	2.60
<i>S_Abs_Accruals_Kothari</i>	800	0.233	0.483	870	0.384	0.769	0.152***	4.79
<i>S_Accruals_Dechow</i>	800	0.041	0.596	870	0.122	0.318	0.082***	3.52
<i>S_Working_Accruals_Dechow</i>	647	-0.008	0.188	716	0.036	0.205	0.044**	4.21
<i>S_MBE</i>	670	0.063	0.243	662	0.089	0.286	0.027*	1.88

Table 3: The Effect of Customers' Blockchain Adoptions on Suppliers' Accrual Earnings Management

This table presents OLS estimates of our baseline regressions with our key and alternative earnings management measures. Panel A presents OLS regression results where the dependent variable is our key earnings management measure, $S_Accruals_Kothari$, which is performance-matched discretionary accruals (Kothari et al. 2005). The variable of interest, C_POST , is an indicator variable that takes a value of one if year t is after a customer firm adopts blockchain, and zero otherwise. Panel B presents the baseline results based on alternative measures. Columns (1)–(3) present OLS regression results, and Column (4) presents probit regression results. In column (1), the dependent variable is the absolute value of Kothari et al. (2005) performance-matched discretionary accruals. In column (2), the dependent variable is the Dechow et al. (1995) modified Jones model for discretionary accruals. In column (3), the dependent variable is the Dechow et al. (1995) modified Jones model for discretionary working capital accruals. In column (4), the dependent variable is MBE , an indicator variable that equals one if the firm is suspected of engaging in meet-or-beat earnings target behavior, and zero otherwise. In both panels A and B, we limit our sample to supplier firms within the two years before and after their customer firms adopt blockchain. Detailed definitions of all variables are provided in Appendix II. Standard errors are clustered by firm and year. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Panel A: Baseline Regression

VARIABLES	$S_Accruals_Kothari$						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
C_POST	0.066** (2.060)	0.079** (2.412)	0.133** (2.418)	0.152*** (3.309)	0.116** (2.063)	0.179** (2.034)	0.165*** (3.440)
S_ROA		0.091 (0.618)	0.051 (0.267)	0.095 (0.649)	0.330*** (4.032)	0.322*** (3.417)	0.127 (0.849)
S_SIZE		−0.015 (−1.607)	−0.003 (−0.281)	−0.015 (−1.602)	−0.102* (−1.943)	−0.118* (−1.938)	−0.017* (−1.801)
S_BM		0.029*** (3.447)	0.002 (0.297)	0.028*** (3.352)	0.032*** (8.817)	0.033*** (10.423)	0.032*** (4.547)
S_LEV		−0.051 (−0.447)	−0.188 (−1.179)	−0.051 (−0.452)	−0.110 (−0.931)	−0.179 (−0.795)	−0.054 (−0.429)
S_RD		−0.001* (−1.780)	−0.001*** (−2.731)	−0.001* (−1.854)	−0.001 (−1.215)	−0.001 (−1.032)	−0.001* (−1.929)
Constant	−0.013 (−0.776)	0.092 (1.187)	−0.057 (−0.135)	0.170 (1.627)	0.748* (1.824)	0.387 (0.897)	0.370 (1.185)
Observations	1,761	1,670	1,215	1,670	1,670	1,670	1,670
Adjusted R-squared	0.002	0.019	0.017	0.020	0.206	0.056	0.052
Customer Control Variables	NO	NO	YES	NO	NO	NO	NO
Firm Fixed Effect	NO	NO	NO	NO	YES	NO	NO
Relationship Fixed Effect	NO	NO	NO	NO	NO	YES	NO
Industry Fixed Effect	NO	NO	NO	NO	NO	NO	YES
Year Fixed Effect	NO	NO	NO	YES	YES	YES	YES

Panel B: Alternative Measures of Earnings Management

VARIABLES	<i>S_Abs_Accruals_</i> <i>Kothari</i> (1) OLS	<i>S_Accruals_</i> <i>Dechow</i> (2) OLS	<i>S_Working_Accruals</i> <i>_Dechow</i> (3) OLS	<i>S_MBE</i> (4) Probit
<i>C_POST</i>	0.184** (2.518)	0.121*** (3.414)	0.048*** (2.597)	0.229** (2.176)
<i>S_ROA</i>	-0.387 (-1.570)	0.088 (0.845)	-0.083 (-1.305)	-0.976** (-2.267)
<i>S_SIZE</i>	-0.038 (-1.465)	-0.016*** (-2.860)	-0.003 (-0.623)	0.010 (0.345)
<i>S_BM</i>	0.061*** (4.762)	-0.035*** (-5.100)	-0.003 (-0.555)	0.048 (0.747)
<i>S_LEV</i>	0.073 (0.537)	0.037 (0.648)	0.091** (2.268)	-0.575* (-1.798)
<i>S_RD</i>	-0.000 (-0.559)	-0.000* (-1.756)	0.009 (0.416)	-0.284 (-1.000)
Constant	0.436 (1.595)	0.463 (1.196)	-0.101* (-1.814)	-1.494*** (-6.004)
Observations	1,670	1,670	1,363	1,332
Adjusted/ Pseudo R-squared	0.154	0.148	0.017	0.051
Industry Fixed Effect	YES	YES	YES	YES
Year Fixed Effect	YES	YES	YES	NO

Table 4: Hype and/or Increased Expectations

This table presents evidence regarding the market expectation and attention changes for supplier firms after customer firms' blockchain adoption. Panel A presents evidence on an event study of suppliers when their customers adopt blockchain. The column labeled "CAR" reports cumulative abnormal stock returns for supplier firms surrounding their customers' blockchain adoptions. Abnormal returns are calculated against benchmarks estimated using a market model or a Fama-French three-factor Model over the 255 trading days ending 46 days before the event date. Market portfolio return is estimated using the CRSP equal-weighted index, and firms must have at least 22 returns during the pre-event estimation window to be included in the analysis. Significance levels are evaluated based on the Patell Z statistics (Patell 1976). Panel B presents results of univariate tests comparing multiple measures of earnings expectations before and after customers adopt blockchain. S_AVG_EST is the average analyst earnings estimate over the two-year period before or after blockchain adoption. S_AVG_REC is average analyst recommendation score: the higher the score, the better the recommendation. $S_AVG_NUM_UP$ is the average number of upward revisions of recommendations, and $S_AVG_NUM_DOWN$ is the average number of downward revisions of recommendations. All variables are defined in Appendix II. *, **, and *** denote significance at the 10, 5, and 1% level, respectively.

Panel A: Stock Price Reaction to Suppliers around Customers' Blockchain Adoption Announcements

Windows	N	CAR	Patell Z
<u>Market Model</u>			
(-5,+5)	575	0.36%***	3.631
(-2,+2)	575	0.17%***	2.948
(-1,+1)	575	0.28%***	3.840
(-2,+1)	575	0.30%***	3.596
(-10,+5)	575	0.48%***	4.041
<u>Fama-French Three-Factor Model</u>			
Windows	N	CAR	Patell Z
(-5,+5)	580	0.16%**	2.155
(-2,+2)	580	0.07%**	2.172
(-1,+1)	580	0.20%***	3.241
(-2,+1)	580	0.25%***	2.932
(-10,+5)	580	0.22%***	3.040

Panel B: Changes in Market Expectation

	Before Adoption			After Adoption			Difference	
	N	Mean	STD	N	Mean	STD	Mean	t-stat
S_AVG_EST	810	1.524	2.355	841	2.343	3.813	0.819***	5.23
S_AVG_REC	563	2.573	0.686	561	2.690	0.655	0.284**	2.82
$S_AVG_NUM_UP$	753	0.124	0.237	850	0.149	0.252	0.0244*	1.99
$S_AVG_NUM_DOWN$	753	0.167	0.337	850	0.131	0.273	-0.0361*	-2.37

Table 5: Reduced Monitoring

This table presents evidence on reduced monitoring for supplier firms after customer firms adopt blockchain. Panel A presents evidence on the changes in investors' information processing effort for suppliers after customers adopt blockchain. We use three measures to capture professional investors' information processing of supplier firms. *S_EDGAR* is the number of visits from the EDGAR database on the earnings announcement date. *S_BLOOMBERG* is the ranking of Bloomberg news searches on the earnings announcement date. These variables are measured on earnings announcement dates during the two years before the customer firm adopts blockchain (Pre-adoption) and during the two years after adoption (Post-adoption). Panel B presents results of the effect of reduced investor information processing on suppliers' earnings management. In each column, we use an indicator variable for whether a firm has decreased EDGAR visits (*S_DES_EDGAR*), or decreased Bloomberg news searches (*S_DES_BLOOMBERG*) during the two years after blockchain adoption compared to the two years before adoption. *POST* is an indicator variable that takes a value of one if year *t* is after a customer firm adopts blockchain, and zero otherwise. The variable of interest in each column is the interaction term between each of these indicator variables with *C_POST*. Detailed definitions of all variables are provided in Appendix II. We limit our sample of supplier firms to the two years before and after their customer firms adopt blockchain. Standard errors are clustered by firm and year. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Panel A: Changes in Information Processing Effort

Measure	Pre-adoption			Post-adoption			Difference			
	N	Mean	Median	N	Mean	Median	Mean	t-stat	Median	Wilcoxon
S_EDGAR	260	87.150	37.000	233	32.665	0.000	-54.480***	-4.44	-37.000***	-10.422
S_BLOOMBERG	2,452	2.131	4	1,254	1.821	0	-0.310***	-4.53	-4***	-4.469

Panel B: Reduced Information Processing Effort and Accrual Earnings Management

VARIABLES	(1)	(2)
	<i>S Accruals Kothari</i>	<i>S Accruals Kothari</i>
<i>S_DES_EDGAR * C_POST</i>	0.183* (1.898)	
<i>S_DES_EDGAR</i>	-0.134** (-2.155)	
<i>S_DES_BLOOMBERG * C_POST</i>		0.126** (1.988)
<i>S_DES_BLOOMBERG</i>		-0.024 (-0.764)
<i>C_POST</i>	0.129 (1.356)	0.091 (1.579)
<i>S_ROA</i>	0.145 (0.707)	-0.053 (-0.189)
<i>S_SIZE</i>	-0.022 (-1.527)	-0.015 (-1.505)
<i>S_BM</i>	0.038*** (7.969)	0.012 (0.193)
<i>S_LEV</i>	-0.154 (-0.691)	0.015 (0.152)
<i>S_RD</i>	-0.434** (-2.579)	-0.093 (-0.657)
<i>Constant</i>	-0.054 (-0.254)	0.365 (1.220)
Observations	588	1,670
Adjusted R-squared	0.194	0.066
Firm Fixed Effect		
Year Fixed Effect	YES	YES
Industry Fixed Effect	YES	YES

Table 6: Alternative Explanation: Efficiency Gain

This table presents OLS regression results of the effect of suppliers' efficiency gains on accrual earnings management. The dependent variable is suppliers' performance-matched discretionary accruals (*S_Accruals_Kothari*). The variable of interest, *POST*, is an indicator variable that takes a value of one if year *t* is after a customer firm adopts blockchain, and zero otherwise. We use five measures to capture operation efficiency: *S_CYC* (operating cycle), *S_INV* (inventory turnover rate), *S_REC* (accounts receivable turnover rate), *S_ATO* (asset turnover rate), and *S_AP* (accounts payable turnover rate). For each variable, we compare the average two years after with two years before blockchain adoption and create dummy variables to indicate whether operational efficiency improves. *S_CYC* takes the value of one if suppliers have reduced operating cycle days after blockchain adoption, and zero otherwise. *S_INV*, *S_REC*, *S_AT*, and *S_AP* each takes the value of one if suppliers have increased their inventory, receivable, asset, and payable turnover rates after blockchain adoption, respectively, and zero otherwise. Detailed definitions of all variables are provided in Appendix II. We limit our sample on supplier firms to the two years before and after their customer firms adopt blockchain. Standard errors are clustered by firm and year. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

VARIABLES	<i>S_Accruals_Kothari</i>				
	(1)	(2)	(3)	(4)	(5)
<i>C_POST</i>	0.114** (2.159)	0.152*** (2.767)	0.149*** (2.890)	0.162** (2.572)	0.167** (2.528)
<i>S_CYC</i>	-0.040 (-1.115)				
<i>S_CYC*C_POST</i>	0.093 (1.451)				
<i>S_INV</i>		-0.009 (-0.232)			
<i>S_INV*C_POST</i>		0.017 (0.251)			
<i>S_REC</i>			-0.061 (-1.508)		
<i>S_REC*C_POST</i>			0.039 (0.584)		
<i>S_ATO</i>				-0.007 (-0.219)	
<i>S_ATO*C_POST</i>				-0.006 (-0.080)	
<i>S_AP</i>					-0.043 (-1.203)
<i>S_AP*C_POST</i>					-0.003 (-0.044)
<i>S_ROA</i>	0.123 (0.796)	0.125 (0.795)	0.124 (0.791)	0.125 (0.799)	0.130 (0.839)
<i>S_SIZE</i>	-0.018* (-1.774)	-0.018* (-1.784)	-0.019* (-1.861)	-0.018* (-1.846)	-0.019* (-1.924)
<i>S_BM</i>	0.031*** (4.743)	0.031*** (4.614)	0.031*** (4.611)	0.031*** (4.618)	0.031*** (4.547)
<i>S_LEV</i>	-0.041 (-0.336)	-0.041 (-0.339)	-0.035 (-0.289)	-0.040 (-0.333)	-0.040 (-0.328)
<i>S_RD</i>	-0.001* (-1.892)	-0.001* (-1.957)	-0.001* (-1.899)	-0.001** (-1.974)	-0.001* (-1.889)
<i>Constant</i>	0.356 (1.136)	0.368 (1.168)	0.376 (1.207)	0.376 (1.196)	0.397 (1.259)
Observations	1,672	1,672	1,672	1,672	1,672
Adjusted R-squared	0.065	0.063	0.064	0.063	0.064

Industry	YES	YES	YES	YES	YES
Year Fixed Effect	YES	YES	YES	YES	YES

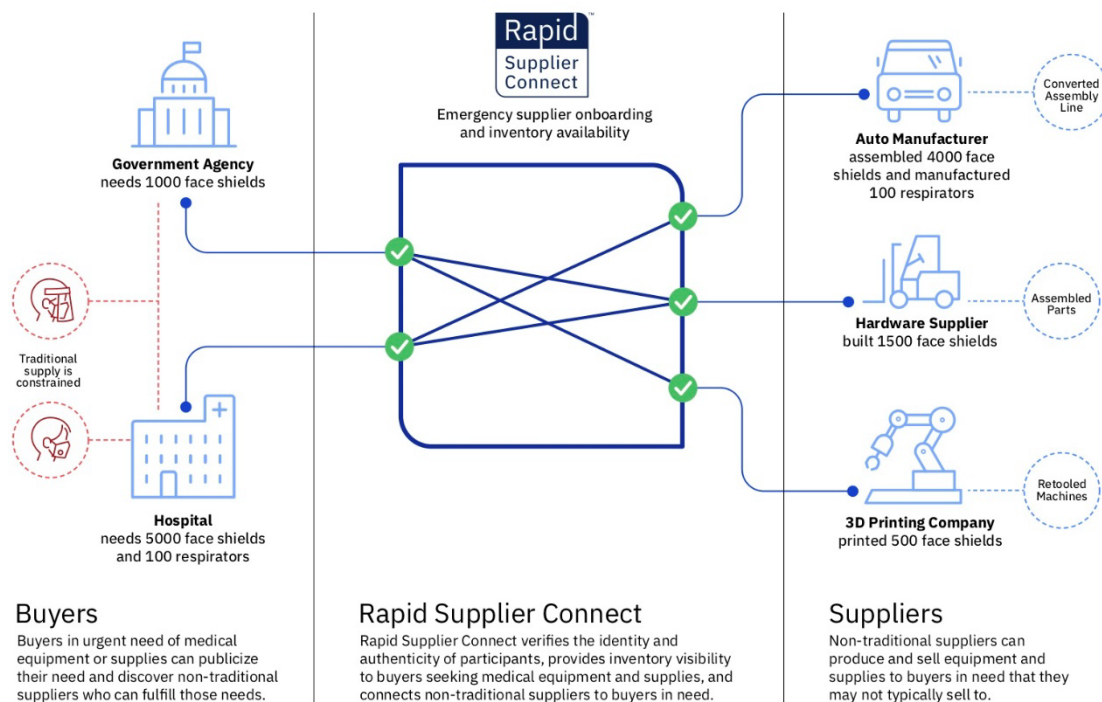
Internet Appendix

Internet Appendix I. Case Study: Application of Blockchain in the Supply Chain

Summary:

Rapid Supplier Connect, launched by IBM, is a blockchain-based network that helps healthcare organizations, whether hospitals or government agencies, address equipment shortages during the pandemic by helping them find alternative suppliers.²⁸ The adoption of this blockchain platform by customers (hospitals and government agencies) has significant impacts on their suppliers. The

Rapid Supplier Connect: Helping Address Medical Supply Chain Shortages



IBM Blockchain

IBM

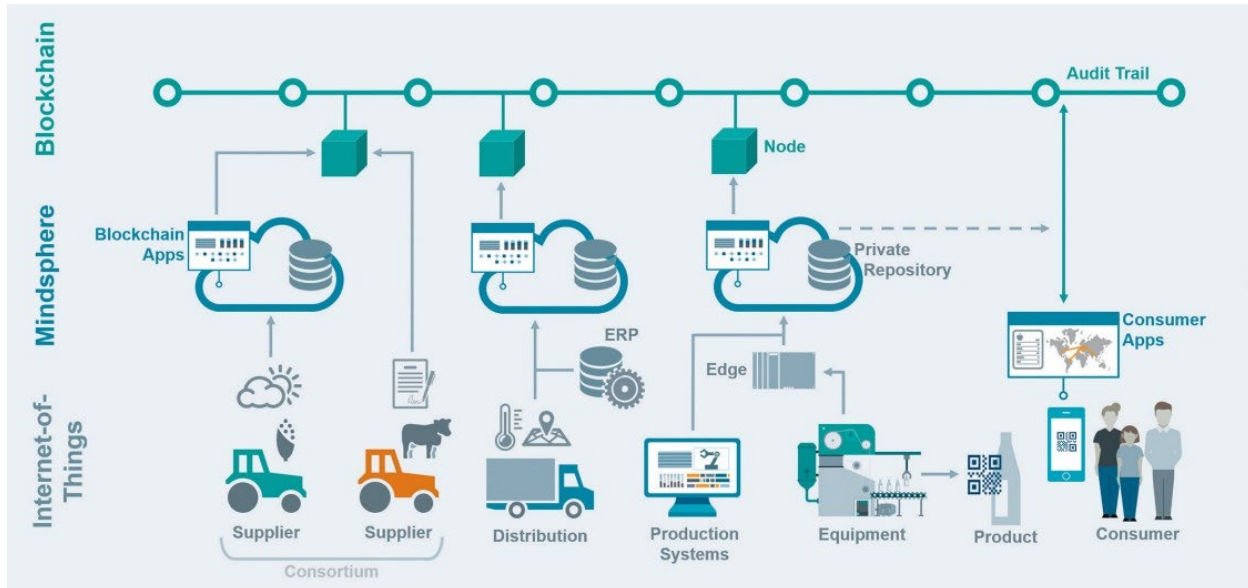
graph below illustrates the network.

²⁸ The case study is a summary of information from these sources:

1. IBM: <https://www.ibm.com/blockchain/solutions/rapid-supplier-connect>
<https://www.trustyoursupplier.com/rapid-supplier-connect/>
2. PRNewsWire: <https://www.prnewswire.com/news-releases/ibm-helping-to-battle-covid-19-medical-supply-chain-shortages-with-the-launch-of-ibm-rapid-supplier-connect-301047295.html>

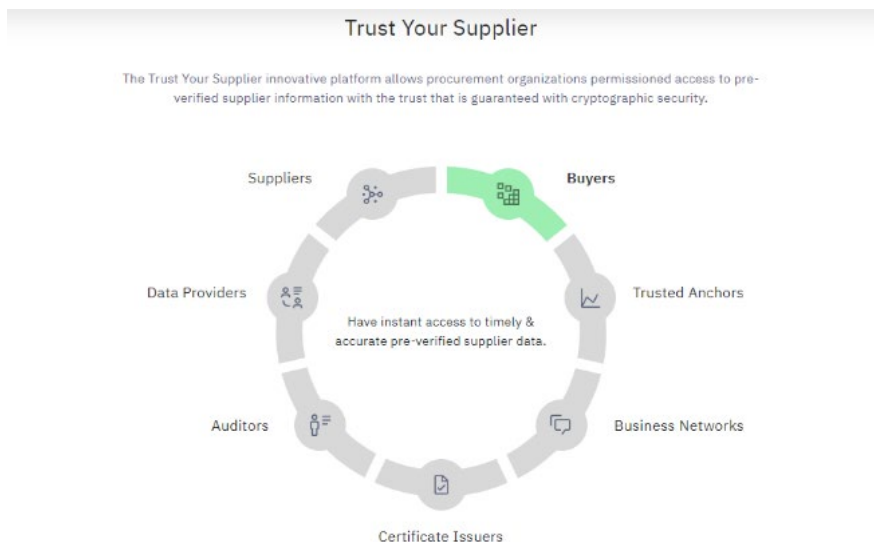
Process:

The network provides validation checks and inventory information made available in near-real time. The network makes it possible for buyers and suppliers—including nontraditional suppliers joining the fight—to quickly find each other, accelerate verification and onboarding processes, and gain near real-time insights into inventories of life-saving equipment. Suppliers' data are tamper-proof and transparent to permissioned customers. The chart below illustrates the data flow through the blockchain.



Technology:

The Rapid Supplier Connect network uses the blockchain-based *Trust Your Supplier* identity platform, which provides a trusted source of supplier information and digital identity that reduces risk while simplifying and accelerating supplier onboarding and supplier management. The chart below illustrates the parties involved in the technology.



Impact:

- **Timeliness:** It provides an efficient supplier onboarding process, taking just 30 minutes to join the network. Inventory information is shared in real time.
- **Transparency:** Buyers can identify new suppliers from a broader range of suppliers in the network. Suppliers also have access to user feedback.
- **Provenance:** Supplier data are tamperproof and made available to permissioned users.
- **Security:** Suppliers have portable online identities and the ability to post and manage inventory availability.
- **Free of Charge:** The network is available at no cost to qualified buyers and suppliers.

Internet Appendix II. Identification Tests: Propensity Score Matching and Entropy Balancing

Although a customer's blockchain adoption is an exogenous decision to the supplier, it can be argued that the blockchain-adopted customers select suppliers with certain firm financial reporting characteristics. To address this concern as well as to minimize the concern of functional form misspecification (FFM) (Shipman et al. 2017), we use propensity score matching by including additional variables and implement difference-in-difference estimations. We also use entropy balancing that reweights the control sample observations without causing significant sample attributions (Hainmueller 2012; Hainmueller and Xu 2013).

In this internet appendix, we describe the procedure of constructing the two propensity score matching samples and the Entropy balancing sample, as well as the difference-in-difference (DiD) test based on these samples. Results are presented and discussed.

Internet Appendix II.1 Propensity Score Matching

Internet Appendix II.1.1 Matching on Customer Characteristics and the DiD Test

To conduct matching, we first interview a director at the Boston Blockchain Association and read several industry reports to gather insights on the determinants of a firm's blockchain adoption decision as no prior study provides guidance. Based on the research, we match firms with characteristics in four dimensions—firm fundamentals, firm growth, industry competition, and information quality—that are likely influence blockchain adoption decisions. We conduct two sets of propensity score matching (PSM), based on customer firm characteristics, and a combination of customer and supplier characteristics, respectively. Next, we conduct a difference-in-difference analysis based on the matched sample.

We identify eight measures that demonstrate significant differences between our sample of blockchain adopting firms versus all other public firms that meet our sample criteria. Panel A of Internet Appendix Table 1 reports the univariate test results. These statistics are gathered in the year before blockchain adoption for both the blockchain-adopting firms (to capture the pre-adoption determinants) and the rest of the public firms. The statistics reveal that blockchain-adopting firms have higher profitability (C_ROA), higher market value ($C_LN(MV)$), lower leverage (C_LEV), higher research intensity (C_RD), and lower capital expenditure growth (C_CAPX_G); are in industries with greater competition (C_HHI); and are more timely in announcing earnings (C_ANN_DAYS). We use these variables as determinants of customers' blockchain adoptions to estimate propensity scores. We also include customer firms' discretionary accruals ($C_ACCRUALS$) since customers and suppliers may display similar accrual earnings management behavior. These variables are defined in Appendix II.

The pool of candidate matching firms for our PSM analysis includes firms that have not adopted blockchain technology in our period of study and for which supplier information is available in FactSet. This pool consists of 8,340 firm-year observations. Following recent studies (Shipman et al. 2017), we use PSM to match each treatment firm with a control firm from the pool by the closest propensity score in year $t-1$, with the blockchain adoption year as year t . Ideally, the matched firms should be the same as the treatment firms in all firm characteristics except for their blockchain adoption status.

The probit model used to perform PSM includes the identified firm characteristics that could potentially affect blockchain adoption decisions, lagged by one year. We use the predicted value of the probit model as the propensity score and match each blockchain-adopting customer firm with a control customer firm, without replacement, by the closest proximity of propensity

score in the same year and in the same industry. We set the caliper width to 3% to ensure the proximity of firm characteristics between treatment firms and matched firms. With this caliper width, we identify 26 matched pairs through PSM.

In Panel B of Internet Appendix Table 1, we compare the treatment and matched firms. None of the variables are statistically different between groups, indicating balanced treatment and matched samples. We construct the supplier firm sample as follows. First, we include all suppliers that are linked with the treatment and matched customer firms. Second, we include two preceding and succeeding firm-year observations for each treatment and matched supplier firm. We use this matched supplier firm sample to conduct our difference-in-difference analyses below.

Using the matched customer firm-pairs, we construct the control and treatment *supplier firm* sample by including all suppliers that are linked with the treatment and matched customer firms, during the two preceding and two succeeding firm-year observations for each supplier firm. Intuitively, the supplier firms linked to the blockchain-adopting customer firms are the treatment supplier firms, and those linked to the non-blockchain-adopting customer firms are the control supplier firms. We ensure the control supplier firms have no customers who have adopted blockchain. This procedure produces our matched supplier firm sample to conduct our DiD analyses.

We use the following difference-in-difference model to capture the causal effect of customer firms' blockchain adoption on supplier firms' earnings management:

$$\begin{aligned}
 S_Accruals_Kothari &= \beta_0 + \beta_1 S_TREAT \\
 &+ \beta_2 C_POST_{i,t} + \beta_3 TREAT * C_POST_{i,t} + \beta_4 Control\ Variables \\
 &+ Industry\ Dummies + Year\ Dummies + \epsilon_{i,t},
 \end{aligned} \tag{1}$$

where $S_Accruals_Kothari$ is our measure of discretionary accrual earnings management. S_TREAT indicates whether a firm is in the treatment group (suppliers with blockchain customers) or control group (suppliers with no blockchain customers), and C_POST is the indicator variable reflecting the post-adoption period. We include all control variables from the baseline regression model (Table 3) and use the four-year event window $[-2, +2]$ for this difference-in-difference model. We include the same set of control variables in this difference-in-difference model as in the baseline models.

Our variable of interest is the interaction term $C_POST*TREAT$. A positive (negative) coefficient on this interaction term would indicate that customers' blockchain adoptions lead to increases (decreases) in accrual earnings management in their supplier firms. Panel C of Internet Appendix Table 1 reports the results. The coefficient of C_POST*S_TREAT in each specification is significant and positive. For example, in Column (3), which includes both industry and year fixed effects, the coefficient on C_POST*S_TREAT is 0.023 (t-stat = 2.053), suggesting that suppliers' accrual earnings management significantly increases after customers adopt blockchain, compared to matched suppliers not affiliated with blockchain customers.

Internet Appendix II.1.2 Matching on Both Customer and Supplier Characteristics, and the DiD Test

Next, we conduct an alternative PSM procedure that incorporates both the supplier firm characteristics and the customer firm characteristics. We create a pool of candidates comprising supplier firms with no blockchain-adopting customers, given the available information in FactSet, leading to 77,347 supplier firm-year observations. Next, we follow recent studies (Shipman et al. 2017) and use PSM to match one control firm to each treatment firm. The matched firms ideally have the same firm characteristics except that only one of the firms in each pair has adopted

blockchain. Each treatment firm is paired with the non-blockchain control firm with the closest propensity score in event year $t-1$.

To obtain the propensity score, we first estimate a probit model with all the supplier firm characteristics included in our baseline regression (i.e., S_ROA , S_SIZE , S_BM , S_LEV , S_RD), the supplier firms' discretionary accruals ($S_ACCRUALS$), the length of the supply chain relationship (S_SUP_DAY), and all the customer characteristics identified in Appendix A.1. All independent variables in the probit model are lagged by one year. Next, we obtain the predicted value of the probit model as the propensity score. Without replacement, we match each supplier firm with the control supplier firm with the closest propensity score in the same year and the same industry. By matching within a caliper of 3%, we identify 247 treatment-match supplier pairs. We compare the statistics of the treatment and matched supplier firms (Panel A of Internet Appendix Table 2). For all variables, the t-statistics for the mean differences between the treatment firms and the matched firms are not statistically significant. The results indicate that our matching method generates balanced treatment and matched samples. After matching, we include the two preceding and succeeding firm-year observations for each treatment and matched supplier firm. We use this matched sample to conduct our difference-in-difference analyses.

Panel B of Internet Appendix Table 2 presents the DiD estimates using Equation (4) and including all PSM variables as control variables, for the period of two years before and two years after the customer firm's blockchain adoption. The variable of interest, $C_POST*TREAT$, is significant and positive in all specifications. For example, in Column (3), which includes both industry and year fixed effects, the coefficient on C_POST*S_TREAT is 0.146 (t-stat = 2.045), consistent with findings in the previous difference-in-difference model.

Internet Appendix II.2 Entropy Balancing

Following prior studies (Hainmueller 2012; Hainmueller and Xu 2013), we balance the treatment and control samples on three sample moments (mean, variance, and skewness) for all covariates in the baseline regression. We reweight the baseline regression using the balanced sample, which has 19,919 observations. In an untabulated test, the coefficient on *C_POST* is 0.083 (t-stat 2.668), significant at the 1% level, consistent with our baseline findings.

Internet Appendix II.3 Placebo Test

Moreover, we conduct a placebo test in which we replace our sample customer firms (who are at an advanced stage of blockchain adoption) with a new sample of firms at a preliminary stage of blockchain adoption. These customer firms are at the research and development stage of blockchain technology or are initiating tests such as a pilot program using blockchain. Market participants would reasonably have lower expectations and less perception of improved financial reporting quality in these blockchain adoptions, as these firms have not yet implemented blockchain. We include suppliers connected to customers at the preliminary stage of adopting blockchain and re-estimate our baseline regressions. Internet Appendix Table 3 presents the results. In models that include controls, the coefficient of *C_POST* is not statistically significant. Therefore, there is no evidence of increased (or decreased) earnings management for preliminary stage adoptions. Taken together, the results reported in this Internet Appendix support our baseline result that a customer firm's blockchain adoption leads to its supplier firms' earnings management.

Internet Appendix III: Robustness Tests for the Baseline Results

Internet Appendix III.1 Balanced Sample

We re-estimate our baseline model using a smaller but balanced sample in which each supplier firm enters during each of the four years surrounding the event date. The results are similar to the baseline estimations.

Internet Appendix III.2 Multiple Customer Firms and Significance of Customer Firms

We conduct tests showing that our results are not affected by whether suppliers have multiple customers adopting blockchain. We separate the sample into first-time customer blockchain adoptions and all other adoptions and repeat our baseline regression estimation. The subsample of first-time adopters contains 1,034 observations, while the subsample of all other adoptions has 630 observations. In both subsample tests, the coefficients of *POST* are positive and significant at the 5% level.

We also control for the significance of the customer firm's revenue to the supplier firm. The SEC requires public companies to disclose any customers with 10% or more of all the revenues.²⁹ To capture the significance of these major customers, we create an indicator variable, *REV_SIG*, which takes the value of one if the percentage of revenue from that customer is 10% or more, and zero otherwise.³⁰ Using this indicator variable, first, we include it as a control variable in the baseline regression model. The main result holds. Second, we interact it with the variable of interest *POST* and find no significant effect of the interaction term. Thus, major customers do not have significantly different influence than non-major customers on suppliers in adopting blockchain.

²⁹ For details of the requirement, please refer to: <https://www.sec.gov/rules/proposed/33-7549.htm>

³⁰ If one customer firm has missing data of revenue percentage, we assume the percentage is less than 10%.

Internet Appendix III.3 Sample Distribution Issue

In the baseline regression, the estimators may be subject to finite sample properties, as we have only 1,670 observations. One may question the sample distribution and the consistency of estimators in the regressions. To alleviate this concern, we first plot the distribution of the baseline regression residuals. The distribution is close to a normal distribution and has a mean value of zero. We notice a clustering around the mean zero, but such clustering may be driven by our dependent variable, which is the estimated residual in Equation (1) and adjusted with performance matching. Second, we conduct a bootstrapping test with 100 replications. Bootstrapping is a test or metric that uses random sampling with replacement. For a dataset containing N observations, bootstrapping draws, with replacement, N observations from the N -observation dataset. In this random drawing, some of the original observations will appear once, some more than once, and some not at all. We use bootstrapping to randomly draw, with replacement, 100 times the baseline regression sample size of 1,670 observations.³¹ The estimators are averaged, and standard errors are estimated on the assumption that the sampling distribution is approximately normal (Hall and Wilson 1991). The results are similar to those produced by the baseline regression.

Internet Appendix III.4 Controlling for the Supply Chain Relationship Materiality

We account for the strength of the customer–supplier relationship by measuring the length of the relationship (which ranges from 23 to 6,298 days). We repeat our baseline estimations using only firm-pairs that have been in the supply chain relationship longer than 365 days (which includes 88% of our sample). In additional tests we control for the relative bargaining power measure developed by Hui et al. (2012), which is the ratio of the average market value of firms in

³¹ A total of 50 to 200 replications are generally adequate for estimates of standard errors and thus are adequate for normal-approximation confidence intervals. MOONEY, C. F.; C. Z. MOONEY; C. L. MOONEY; R. D. DUVAL and R. DUVALL. *Bootstrapping: A nonparametric approach to statistical inference*. sage, 1993.

suppliers' industries to a customer's own market value. In each of these tests the results are consistent with those reported in the baseline estimations.

Internet Appendix III.5 Real Earnings Management

We conduct two tests using the composite measure of real earnings management (REM), following Roychowdhury (2006). REM includes three types of real activities manipulation – sales manipulation, reduction of discretionary expenditures, and overproduction to lower cost of goods sold. First, we control for REM in our baseline regression and find similar results. Second, in our baseline regression we replace accruals earnings management with REM and find that suppliers' real earnings management increases after customers' blockchain adoption, but this increase is due to a reduction of abnormal discretionary expenses.

Internet Appendix III.6 Dynamic Analysis

To address the concern that the increased earnings management is only a temporary response to the market hype of blockchain, we conduct a dynamic analysis using a sample consists of three years before and after customers' blockchain adoptions (i.e., 6 years in total), and by replacing the indicator variable *C_POST* with indicator variables capturing the number of years before or after blockchain adoption. Untabulated results show that, on average, supplier firms' accrual earnings management increases within the two years after a customer firm's blockchain adoption and appears to stay at the same level in the third year after blockchain adoption. The evidence is inconsistent with short-term earnings management followed by accrual reversals.

Internet Appendix III.7 Backward versus Forward Integration

It is possible that blockchains are integrated downstream from suppliers to customers. We use textual analysis to search the keyword “customer” in each of the news articles collected. It's possible that these firms apply blockchain technology to downstream customers rather than

upstream suppliers. We find that 13 firms mention “customer” in the news article. We remove these observations from the baseline regression and find similar results to those we report.

.

Internet Appendix Table 1: Propensity Score Matching on Customer Characteristics

This table reports the endogeneity test, a difference-in-differences test based on treatment and control samples matched by characteristics of customer firms. We match the blockchain adopters with nonblockchain adopters using profitability (C_ROA), market value ($C_LN(MV)$), leverage (C_LEV), research intensity (C_RD), and capital expenditure growth (C_CAPX_G), industry's competition intensity (C_HHI), earnings announcement speed (C_ANN_DAYS), and discretionary accruals ($C_ACCRUALS$). One-to-one propensity score matching is performed by choosing the closest proximity of propensity scores without replacement, and the caliper is set to 3%. We then use the matched "treatment" and "control" group to identify treatment and control supplier firms — all suppliers related to the "treatment customer" would be treatment supplier firms, and all those related to the "control customer" would be control supplier firms. We remove suppliers that have invested in blockchain technology but that are related to a control customer from the control supplier group to ensure the cleanness of the test. Panel A reports results of univariate tests comparing multiple firm characteristics of blockchain-adopting firms before adoption with the rest of public firms over the sample period. The prefix "C_" indicates the variables are for customer firms. Panel B verifies the post-matching differences between the two groups in these variables. The panel reports results when we compare the means of the PSM-matching variables between the treatment and the matched customer firms. Panel C presents difference-in-differences OLS regression results based on the supplier firms. The dependent variable is the Kothari et al. (2005) performance-matched discretionary accruals. The variable C_POST is an indicator variable that takes a value of one if year t is after a customer firm adopts blockchain, and zero otherwise. S_TREAT equals one if a supplier firm is in the treatment group, and zero if otherwise. Our variable of interest is the interaction term, $C_POST*TREAT$. We limit our sample to supplier firm-year observations within the two years before and two years after their customer firms adopt blockchain. Standard errors are clustered by firm and year. Detailed definitions of all variables are provided in Appendix II. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Panel A: Firm Characteristics of Blockchain Adopters

	Blockchain-Adopting Firms			Other Public Firms			Mean Difference	t-stat	Wilcoxon Significance
	Mean	Median	Std	Mean	Median	Std			
C_ROA	0.018	0.013	0.193	-0.022	0.014	0.199	0.040*	2.53	
$C_LN(MV)$	7.870	8.581	3.058	5.365	5.389	2.628	2.505***	11.34	***
C_LEV	0.172	0.111	0.183	0.267	0.202	0.280	-0.095***	-3.88	***
C_RD	0.084	0.000	0.126	0.020	0.000	0.070	0.063***	11.70	***
C_CAPX_G	0.026	0.000	0.577	0.663	0.000	3.027	-0.637**	-1.97	
C_HHI	0.332	0.226	0.283	0.228	0.113	0.254	0.104***	4.08	***
C_ANN_DAYS	39.956	36.000	19.349	49.876	49.000	22.765	-9.921***	-5.05	***
$C_ACCRUALS$	-0.063	-0.003	1.078	0.016	0.000	2.043	-0.0796	-0.86	

Panel B: Post-Matching Sample Difference for Customers' Covariates

	Treatment Sample	Matched Sample	Difference	
	Mean	Mean	Mean	t-stat
C_ROA	0.019	0.059	-1.17	0.247
$C_LN(MV)$	9.024	8.951	0.12	0.908
C_LEV	0.186	0.170	0.32	0.752
C_RD	0.126	0.123	0.10	0.924
C_CAPX_G	0.027	-0.090	1.36	0.179
C_HHI	5.055	2.782	0.86	0.394
C_ANN_DAYS	38.962	39.231	-0.05	0.959
$C_ACCRUALS$	-0.013	-0.081	1.20	0.235

Panel C: DiD Test

VARIABLES	<i>S_Accruals_Kothari</i>		
	(1)	(2)	(3)
<i>C_POST</i>	−0.000 (−0.004)	−0.003 (−0.454)	−0.004 (−0.470)
<i>TREAT</i>	−0.001 (−0.015)	−0.000 (−0.000)	−0.002 (−0.022)
<i>C_POST*TREAT</i>	0.023* (1.996)	0.026* (1.918)	0.023** (2.053)
<i>S_ROA</i>		−0.003 (−0.825)	−0.003 (−0.848)
<i>S_SIZE</i>		0.006 (1.455)	0.007* (1.890)
<i>S_BM</i>		−0.022 (−1.373)	−0.026* (−1.707)
<i>S_LEV</i>		0.000 (0.770)	−0.000 (−0.255)
<i>S_RD</i>		−0.000 (−0.239)	0.000 (0.870)
Constant	−0.060 (−1.605)	−0.099* (−1.776)	0.289 (1.532)
Observations	2,452	2,409	2,409
Adjusted R-squared	0.002	0.009	0.232
Industry Fixed Effect	NO	NO	YES
Year Fixed Effect	NO	NO	YES

Internet Appendix Table 2: Propensity Score Matching on Both Customer and Supplier Characteristics

This table reports the endogeneity test, a difference-in-differences test based on treatment and control samples matched by characteristics of the customer and supplier firm. The treatment supplier firms are firms with one or more customers who adopted blockchain during the sample period, and the control supplier firms are matched to the treatment firms from a pool of firms without any customers who adopted blockchain. We match the supplier firms using the following characteristics of the customer firm (denoted by a prefix “C_”): profitability (C_ROA), market value ($C_LN(MV)$), leverage (C_LEV), research intensity (C_RD), capital expenditure growth (C_CAPX_G), industry’s competition intensity (C_HHI), earnings announcement speed (C_ANN_DAYS), and discretionary accruals ($C_ACCRUALS$), as well as the following characteristics of the supplier firms: discretionary accruals ($S_ACCRUALS$), S_ROA , S_SIZE , book-to-market ratio (S_BM), Leverage (S_LEV), R&D intensity (S_RD), and the length of the supply chain relationship (S_SUP_DAY). One-to-one PSM matching is performed by choosing the closest proximity of propensity scores without replacement, and the caliper is set to 3%. Panel A reports the post-matching differences in these variables for the treatment and control supplier firm groups. Panel B presents difference-in-differences OLS regression results based on the supplier firms. The dependent variable is the Kothari et al. (2005) performance-matched discretionary accruals. The variable C_POST is an indicator variable that takes a value of one if year t is after a customer firm adopts blockchain, and zero otherwise. S_TREAT equals one if a supplier firm is in the treatment group, and zero if otherwise. Our variable of interest is the interaction term, $C_POST*TREAT$. Standard errors are clustered by firm and year. We limit our sample to supplier firm-year observations within the two years before and two years after their customer firms adopt blockchain. Detailed definitions of all variables are provided in Appendix II. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Panel A: Post-Matching Sample Difference for Supplier and Customer Covariates

	Treatment Sample	Matched Sample	Difference	
	Mean	Mean	Mean	t-stat
$S_ACCRUALS$	−0.059	−0.034	−0.025	−0.69
S_ROA	0.060	0.036	0.024	1.38
S_SIZE	7.657	7.650	0.006	0.03
S_BM	0.396	0.368	0.028	0.77
S_LEV	0.179	0.174	0.004	0.24
S_RD	0.178	0.172	0.006	0.26
S_SUP_DAY	1405.900	1318.800	87.100	0.84
C_ROA	0.117	0.116	0.001	0.10
$C_LN(MV)$	10.172	10.173	−0.001	0.01
C_LEV	0.171	0.186	−0.015	−1.01
C_RD	0.109	0.115	−0.006	−0.54
C_CAPX_GROWTH	82.436	64.597	17.839	0.58
C_HHI	19.302	18.303	0.999	0.45
C_ANN_DAYS	−34.147	−14.329	−19.818	−1.17
$C_ACCRUALS$	0.129	0.138	−0.009	−0.23

Panel B: Difference-in-Differences Test

VARIABLES	<i>S_Accruals_Kothari</i>		
	(1)	(2)	(3)
<i>C_POST</i>	−0.080 (−1.426)	−0.071 (−1.262)	−0.026 (−0.294)
<i>TREAT</i>	−0.041 (−1.150)	−0.039 (−1.100)	−0.035 (−0.971)
<i>C_POST*TREAT</i>	0.171** (2.450)	0.159** (2.310)	0.146** (2.045)
<i>S_ROA</i>		0.240*** (3.451)	0.247*** (3.414)
<i>S_SIZE</i>		−0.020** (−2.387)	−0.020** (−2.120)
<i>S_BM</i>		−0.013 (−0.299)	−0.005 (−0.105)
<i>S_LEV</i>		−0.023 (−0.488)	−0.033 (−0.748)
<i>S_RD</i>		−0.009 (−1.329)	−0.008 (−1.346)
<i>S_SUP_DAY</i>		0.000 (0.686)	0.000 (0.698)
<i>C_ROA</i>		0.006** (2.532)	0.008* (1.852)
<i>C_LN(MV)</i>		0.010 (1.118)	0.008 (0.850)
<i>C_LEV</i>		0.042 (0.684)	0.040 (0.600)
<i>C_RD</i>		0.171 (0.894)	0.155 (0.751)
<i>C_CAPX_GROWTH</i>		0.000* (1.781)	0.000* (1.721)
<i>C_HHI</i>		−0.001 (−0.810)	−0.001 (−0.722)
<i>C_ANN_DAYS</i>		0.000 (0.282)	0.000 (0.177)
<i>C_Accruals</i>		−0.027 (−1.032)	−0.022 (−0.823)
Constant	0.005 (0.156)	0.056 (0.490)	−0.075 (−0.178)
Observations	1,137	1,137	1,137
Adjusted R-squared	0.005	0.025	0.008
Industry Fixed Effect	NO	NO	YES
Year Fixed Effect	NO	NO	YES

Internet Appendix Table 3: Placebo Test - Customer Firms at the Preliminary Stage of Adopting Blockchain

This table presents OLS regression results of the placebo test. The sample comprises supplier firms that are connected to customers who are in the preliminary stages of adopting blockchain. The dependent variable is Kothari et al. (2005) performance-matched discretionary accruals. The variable of interest, *C_POST*, is an indicator variable that takes a value of one if year *t* is after a customer firm adopts blockchain, and zero otherwise. Detailed definitions of all variables are provided in Appendix II. We limit our sample to supplier firms within the two years before and after their customer firms adopt blockchain. Standard errors are clustered by firm and year. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

VARIABLES	<i>S_Accruals_Kothari</i>				
	(1)	(2)	(3)	(4)	(5)
<i>C_POST</i>	0.038** (2.171)	0.015 (0.637)	-0.020 (-1.137)	0.021 (0.577)	-0.022 (-1.094)
<i>S_ROA</i>		0.075 (0.948)	0.153* (1.827)	0.139 (1.475)	0.121* (1.830)
<i>S_SIZE</i>		-0.015*** (-2.655)	-0.038 (-1.184)	-0.033 (-0.831)	-0.019*** (-3.090)
<i>S_BM</i>		-0.000 (-1.499)	-0.000 (-1.274)	-0.000 (-1.210)	-0.000 (-1.586)
<i>S_LEV</i>		-0.079* (-1.650)	-0.076 (-1.055)	-0.098 (-0.926)	-0.052 (-1.069)
<i>S_RD</i>		0.000 (1.219)	0.000 (0.990)	0.000 (0.901)	0.000 (1.171)
Constant	-0.030** (-2.393)	0.345** (2.412)	0.622** (2.306)	0.367 (0.176)	0.394*** (3.359)
Observations	10,951	8,502	10,297	10,297	10,297
Adjusted R-squared	0.001	0.007	0.197	0.051	0.019
Customer Control Variables	NO	YES	NO	NO	NO
Firm Fixed Effect	NO	NO	YES	NO	NO
Relationship Fixed Effect	NO	NO	NO	YES	NO
Industry Fixed Effect	NO	NO	NO	NO	YES
Year Fixed Effect	NO	YES	YES	YES	YES

Internet Appendix Table 4. Purposes of Blockchain Adoption

This table presents OLS regression results. Each column includes subsamples for a specific purpose of adopting blockchain. The dependent variable is Kothari et al. (2005) performance-matched discretionary accruals. The variable of interest, *C_POST*, is an indicator variable that takes a value of one if year *t* is after a customer firm adopts blockchain, and zero otherwise. Detailed definitions of all variables are provided in Appendix II. We limit our sample on supplier firms to the two years before and after their customer firms adopt blockchain. Standard errors are clustered by firm and year. *, **, and *** denote significance at the 10, 5, and 1% level, respectively.

VARIABLES	<i>S_Accruals_Kothari</i>			
	(1)	(2)	(3)	(4)
	<i>Financial</i>	<i>Nonfinancial</i>	<i>Nonfinancial- Advantage</i>	<i>Nonfinancial- Information</i>
<i>C_POST</i>	0.139 (1.028)	0.196*** (3.035)	0.206** (2.222)	0.158 (1.528)
<i>S_ROA</i>	-0.216 (-0.677)	0.266*** (2.767)	-0.491 (-1.392)	0.349*** (5.129)
<i>S_SIZE</i>	-0.014 (-1.059)	-0.017 (-1.362)	0.025 (1.333)	-0.041* (-1.777)
<i>S_BM</i>	0.039*** (7.536)	0.002 (0.287)	-0.007 (-1.013)	0.008 (0.300)
<i>S_LEV</i>	-0.099 (-0.515)	-0.068 (-0.604)	0.093 (0.609)	-0.125 (-0.770)
<i>S_RD</i>	-0.001 (-1.619)	-0.002*** (-3.175)	-0.091 (-0.630)	-0.002*** (-3.627)
Constant	0.255 (1.377)	0.388 (1.197)	-0.194 (-0.939)	0.352 (0.877)
Observations	867	778	358	420
Adjusted R-squared	0.102	0.007	0.065	0.017
Industry Fixed Effect	YES	YES	YES	YES
Year Fixed Effect	YES	YES	YES	YES